EPFL

An Asymptotic-Preserving Scheme for Braginskii Equations

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Abstract

This work focuses on the development of asymptotic-preserving finite element methods for reduced plasma models derived from the two-fluid Braginskii equations. The objective is to construct robust numerical schemes for a reduced electromagnetic subsystem in magnetized plasma regimes characterized by strong anisotropies and multiscale behavior.

The models involve singular perturbation structures due to small physical parameters, which can lead to stability and accuracy issues for classical numerical methods. To address this, an asymptotic preserving method was created based on the method used by Pierre Degond in [6].

The numerical framework is implemented in the MFEM library and is progressively validated through a hierarchy of model problems, including stationary and time-dependent cases. Finite element discretizations, mixed formulations, and saddle-point problems with Lagrange multipliers are studied, together with implicit time integration schemes such as backward Euler and Crank–Nicolson.

The final formulation of the reduced electromagnetic model leads to a mixed finite element system, whose numerical results confirm the robustness of the method in the asymptotic regime.

Résumé

Ce travail porte sur le développement de méthodes d'éléments finis préservant l'asymptotique (asymptotic-preserving) pour des modèles réduits de plasma dérivés des équations bi-fluides de Braginskii. L'objectif est de construire des schémas numériques robustes pour un sous-système électromagnétique dans des régimes de plasma magnétisé caractérisés par de fortes anisotropies et un comportement multi-échelle.

Les modèles considérés présentent des structures de perturbation singulière dues à la présence de petits paramètres physiques, ce qui peut entraîner des problèmes de stabilité et de précision pour les méthodes numériques classiques. Afin de remédier à ces difficultés, une méthode préservant l'asymptotique a été développée à partir de celle proposée par Pierre Degond dans [6].

Le cadre numérique est implémenté dans la bibliothèque MFEM et est validé progressivement au moyen d'une hiérarchie de problèmes modèles, comprenant des cas stationnaires et dépendant du temps. Les discrétisations par éléments finis, les formulations mixtes ainsi que les problèmes de point selle faisant intervenir des multiplicateurs de Lagrange sont étudiés, de même que des schémas d'intégration temporelle implicites tels que ceux d'Euler et de Crank–Nicolson.

La formulation finale du modèle électromagnétique réduit conduit à un système mixte d'éléments finis, dont les résultats numériques confirment la robustesse de la méthode dans le régime asymptotique.

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1 Introduction

Plasma physics models typically involve the interaction of several physical processes occurring over a wide range of spatial and temporal scales. Among the models used to describe magnetized plasmas, the two-fluid Braginskii equations offer a detailed macroscopic description by treating ions and electrons as two coupled fluids[8]. While these equations are able to capture a rich variety of physical phenomena, their numerical simulation is challenging due to strong anisotropies and stiff terms induced by the presence of a magnetic field.

In many relevant physical regimes, certain parameters become very small, which leads to singular perturbation problems. In such situations, standard numerical methods may face severe stability constraints or lose accuracy when approaching asymptotic limits. To address these issues, asymptotic-preserving (AP) methods have been developed. The main idea behind these methods is to remain stable and accurate independently of the small parameter, while still recovering the correct limiting model as the parameter tends to zero.

This work is motivated by the development of an asymptotic-preserving finite element discretization for a reduced electromagnetic subsystem derived from the two-fluid Braginskii equations. The long-term goal is to implement such methods within the finite element library MFEM, a modern framework designed for high-order finite element computations and large-scale scientific simulations.

Before tackling the full electromagnetic model, it was necessary to develop a solid understanding of the mathematical and numerical tools required for its discretization. For this reason, a significant part of this thesis is devoted to finite element spaces and mixed finite element formulations.

Since the model under study is time-dependent, suitable temporal discretization methods are required. The thesis therefore reviews implicit Runge-Kutta methods, with particular emphasis on the implicit Euler and Crank-Nicolson schemes, which are used throughout the numerical experiments. In addition, the final formulations naturally lead to saddle-point problems involving constraints and Lagrange multipliers. For this reason, the concept of inf-sup stability is introduced to provide the mathematical background needed for the analysis of mixed formulations.

A progressive validation strategy is then adopted. Several model problems of increasing complexity are implemented and solved within MFEM. These test cases serve to verify the correctness of the finite element implementation, validate the assembly of mixed operators, confirm expected convergence properties, and provide practical experiences with the software framework. Starting from simple stationary problems, the study gradually introduces time dependence, coupling effects, and more complex operators, eventually reaching systems that closely resemble the target electromagnetic model.

Building on these preliminary developments, the reduced electromagnetic problem is reformulated using Lagrange multipliers to enforce constraints arising from the decomposition of the solution into components parallel and perpendicular to the magnetic field. This leads to a mixed finite element system that can be implemented using the tools developed in the previous chapters. Numerical experiments are then carried out to study convergence properties and to investigate the behavior of the method for decreasing values of the asymptotic parameter.

The results obtained in this work demonstrate a successful implementation of the proposed formulation within MFEM and provide encouraging evidence of its robustness in the asymptotic regime. While a complete theoretical analysis and the construction of efficient preconditioners remain topics for future work, the present developments establish a solid foundation for the design of asymptotic-preserving finite element methods for plasma models derived from the Braginskii equations.

The remainder of this manuscript is organized as follows. Chapter 2 introduces the Braginskii equations and provides physical context for the work. Chapter 3 presents the reduced electromagnetic model considered in this thesis. Chapters 4 to 6 cover the mathematical and numerical background, including finite element spaces, Runge-Kutta methods, and inf-sup stability theory. Chapter 7 describes the sequence of model problems used to validate the implementation in MFEM. Chapter 8 introduces the Lagrange multiplier framework. Chapter 9 presents the final mixed formulation of the reduced electromagnetic system together with numerical results. Finally, Chapter 10 summarizes the main conclusions and discusses possible directions for future research.

2 Two fluid plasma equations

The model considered in this work is derived from the more general two-fluid plasma formulation presented in the thesis of Maurizio Giacomin [8]. In its full form, the model consists of continuity, momentum, and temperature equations for electrons and ions, including collisional effects, ionization and recombination processes. In the present work, we focus on a reduced version of this model that retains the essential mass and momentum dynamics together with the electromagnetic coupling.

The dynamics of a magnetized plasma can be described using a two-fluid formulation, where electrons and ions are treated as two distinct but coupled fluids. This approach makes it possible to capture important physical effects arising from the interaction between charged particles and electromagnetic fields while retaining a fluid description of the plasma. Each species is characterized by macroscopic quantities such as density, velocity, and pressure.

The governing equations consist of a set of coupled partial differential equations expressing conservation of mass and momentum for each species. These equations are strongly influenced by electromagnetic forces, in particular the Lorentz force, which couples the fluid motion to the electric and magnetic fields. In this work, the magnetic field is assumed to be prescribed and time-independent. Under this assumption, Faraday's law:

$$\partial_t B + \nabla \times E = 0,$$

reduces to

$$\nabla \times E = 0.$$

Consequently, in a simply connected domain, the electric field derives from an electrostatic potential ϕ

$$E = -\nabla \phi.$$

In addition to the fluid equations, the model includes a constraint on the charge. This condition ensures charge conservation and provides an additional coupling between the two species. As a result, the electron and ion dynamics are strongly interdependent through electromagnetic effects, leading to a nonlinear coupled system.

Under these assumptions, the reduced two-fluid plasma model for each species $s \in \{e, i\}$ can be written as

$$\begin{cases} \partial_t n_s + \nabla \cdot (n_s u_s) = 0, \qquad s \in \{e, i\}, \\ m_s \left[\partial_t (n_s u_s) + \nabla \cdot \left(n_s u_s \otimes u_s\right)\right] + \nabla p_s - q_s n_s \Big(-\nabla \phi + u_s \times B\Big) = 0, \qquad s \in \{e, i\}, \\ \nabla \cdot j = \displaystyle\sum_{s \in \{e,i\}} \nabla \cdot \left(q_s n_s u_s\right) = 0. \end{cases}$$

The variables appearing in the model are defined as follows:

- The subscript e refers to electrons and i to ions;
- n_s denotes the density of species s;
- u_s denotes the velocity of species s;

- p_s denotes the pressure of species s;
- m_s denotes the mass of species s;
- q_s denotes the electric charge of species s;
- B denotes the magnetic field;
- ϕ denotes the electrostatic potential;
- j denotes the electric current density.

As discussed previously, the system involves a wide range of characteristic time scales. The fastest processes occur on the order of 10^{-12} s, whereas the slower phenomena of interest evolve on time scales of order 10^{-5} s. This pronounced scale separation makes the numerical simulation of the system particularly challenging, since standard explicit methods would require prohibitively small time steps for stability.

The development of numerical methods capable of efficiently handling these multiple scales while accurately capturing the slow dynamics constitutes the main motivation for the work presented in the following sections.

3 Reduced electro-magnetic model

The main difficulty of the full system comes from the fact that the contributions induced by the electromagnetic fields evolve on a significantly faster time scale than those associated with the fluid dynamics. This disparity leads to stiffness in the equations and complicates both the analytical and numerical treatment of the model.

In order to better understand and handle this multiscale behavior, we first isolate the electromagnetic effects by considering a reduced subsystem involving only the corresponding terms. This preliminary step allows us to study the dominant fast dynamics before coupling them back to the fluid part of the system.

More precisely, we consider the following reduced subsystem, which is obtained by setting $n_s = 1$ for all species s and neglecting both the advection and pressure contributions, namely the terms $\nabla \cdot \big(n_s u_s\big)$, $\nabla \cdot \big(n_s u_s \otimes u_s\big)$ and ∇p_s. This simplification allows us to isolate the dominant electromagnetic effects and focus on the fast dynamics induced by the fields

$$\begin{cases} \partial_t u_s - \dfrac{q_s}{m_s}\Big(-\nabla\phi + u_s \times B\Big) = 0, \qquad s \in \{e, i\}, \\ \nabla \cdot j = \displaystyle\sum_{s\in\{e,i\}} \nabla \cdot \big(q_s u_s\big) = 0. \end{cases}$$

We further assume that $q_e = -q_i$, meaning that electrons and ions carry opposite charges. We then define the new parameters ϵ and μ namely the charge to ion mass ratio ϵ and the electron to ion mass ratio μ in the following way:

$$\epsilon = \frac{m_i}{q_i}, \qquad \mu = \frac{m_e}{m_i}.$$

We note that, in typical physical regimes, ϵ is of order 10^{-8} while μ is of order 10^{-3}. In terms of these parameters, the system can be rewritten in the following form:

$$\begin{cases} \partial_t u_i - \dfrac{1}{\epsilon}\Big(-\nabla\phi + u_i \times B\Big) = 0, \\ \partial_t u_e + \dfrac{1}{\epsilon\mu}\Big(-\nabla\phi + u_e \times B\Big) = 0, \\ \nabla \cdot (u_i - u_e) = 0. \end{cases}$$

The objective of this work is to develop and implement a numerical method for the reduced electromagnetic model, initially considering the regime $\epsilon = 1$ and $\mu = 10^{-3}$. This baseline configuration is chosen as a starting point in order to validate the correctness of the formulation and to assess the basic stability and accuracy properties of the proposed approach in a moderately stiff regime. Based on the performance of the method in this setting, we then aim to progressively explore more demanding multiscale regimes by decreasing the parameter ϵ. The ultimate goal is to investigate how the numerical scheme behaves as the system becomes increasingly stiff.

4 Finite element method

In this project, we are required to solve partial differential equations. To this end, we employ the finite element method (FEM). Hence, we first present review of its underlying principles.

4.1 Motivation and General Idea

Many problems in physics, engineering, and applied mathematics are formulated in terms of partial differential equations (PDEs). These equations describe how physical quantities such as temperature, displacement, pressure, or electromagnetic fields vary continuously in space and time. They are typically defined on a spatial domain, together with boundary conditions that encode the interaction with the exterior or prescribed physical constraints. In most realistic settings, however, these equations cannot be solved exactly, except in highly simplified geometries or under strong assumptions on coefficients and data.

The finite element method [2] provides a systematic framework for constructing approximate solutions to such problems. Its main idea is to transform the original PDE, which is inherently infinite-dimensional, into a finite-dimensional problem that can be solved numerically. This is achieved by first rewriting the PDE in a weak (variational) form, where the notion of solution is relaxed: instead of satisfying the equation pointwise, the solution is required to satisfy an integral identity against a class of test functions. This reformulation reduces regularity requirements and provides a mathematically well-posed setting in appropriate function spaces.

Once the weak formulation is established, the next step is to approximate the infinite-dimensional function space by a finite-dimensional subspace. This space is constructed using basis functions that are typically associated with a decomposition of the domain into simple geometric elements, such as triangles in two dimensions or tetrahedra in three dimensions. These elements form a mesh that approximates the geometry of the domain. The unknown solution is then expressed as a linear combination of these basis functions, which reduces the continuous problem to a finite system of algebraic equations.

This sequence: reformulation, discretization of the domain and projection onto a finite-dimensional function space forms the core structure of the finite element method. It provides a flexible and powerful framework capable of handling complex geometries, heterogeneous materials, and a wide range of boundary conditions while maintaining a solid mathematical foundation.

4.2 Review on Hilbert Spaces

Before introducing the weak formulation, we first recall the notion of a Hilbert space [5]. We begin with the definitions of an inner product space and of completeness in a metric space.

Definition 1 (Inner product space)**.** *Let E be a vector space over $\mathbb{C}$. A mapping $\langle \cdot, \cdot \rangle : E \times E \to \mathbb{C}$ is called an inner product on E if for all $x, y, z \in E$ and all $\alpha, \beta \in \mathbb{C}$, the following properties hold*

1. $\langle x, y \rangle = \overline{\langle y, x \rangle}$
2. $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$
3. $\langle x, x \rangle \geq 0$
4. $\langle x, x \rangle = 0 \iff x = 0$

An inner product naturally induces a norm defined by

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

This norm defines a metric on E through

$$d(x, y) = \|x - y\|.$$

We now introduce the notion of completeness through Cauchy sequences [3].

Definition 2 (Cauchy sequence)**.** *Let (X, d) be a metric space. A sequence $(x_n)_{n\geq 1}$ in X is called a* Cauchy sequence *if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$d(x_n, x_m) < \varepsilon, \qquad \forall n, m \geq N.$$

Definition 3 (Complete metric space)**.** *A metric space (X, d) is said to be* complete *if every Cauchy sequence converges to a limit belonging to X.*
We can now define Hilbert spaces.

Definition 4 (Hilbert space)**.** *A Hilbert space is a complete inner product space.*

The Hilbert spaces that will be used throughout this work are the spaces $L^2(\Omega)$ and $H^1(\Omega)$. Let $\Omega \subset \mathbb{R}^d$ be a bounded open domain with Lipschitz boundary $\partial\Omega$.

Definition 5 (L^2 space)**.** *The space $L^2(\Omega)$ is defined by*

$$L^2(\Omega) = \left\{ v : \Omega \to \mathbb{R} \;\middle|\; \int_\Omega |v|^2 \, dx < +\infty \right\}.$$

It is a Hilbert space equipped with the inner product

$$(f, g)_{L^2(\Omega)} = \int_\Omega fg \, dx,$$

and the associated norm

$$\|f\|_{L^2(\Omega)} = \left(\int_\Omega |f|^2 \, dx \right)^{1/2}.$$

Definition 6 (H^1 space)**.** *The Sobolev space $H^1(\Omega)$ is defined as*

$$H^1(\Omega) = \left\{ v \in L^2(\Omega) \;\middle|\; \nabla v \in (L^2(\Omega))^d \right\}.$$

The space $H^1(\Omega)$ is a Hilbert space endowed with the inner product

$$(f, g)_{H^1(\Omega)} = \int_\Omega fg \, dx + \int_\Omega \nabla f \cdot \nabla g \, dx,$$

and the norm

$$\|f\|_{H^1(\Omega)} = \left(\|f\|^2_{L^2(\Omega)} + \|\nabla f\|^2_{L^2(\Omega)} \right)^{1/2}.$$

We also introduce the space

$$H^1_0(\Omega) = \left\{ v \in H^1(\Omega) \;\middle|\; v|_{\partial\Omega} = 0 \right\},$$

which incorporates homogeneous Dirichlet boundary conditions.

These functional spaces provide the natural framework for the weak formulation of partial differential equations.

4.3 Weak formulation

To introduce the concept of weak formulation, we consider the Poisson problem as a model example [2]. Let $\Omega \subset \mathbb{R}^d$ be a bounded domain and consider

$$-\Delta u = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

This is the strong formulation of the problem, meaning that the differential equation is satisfied pointwise and that the solution u is assumed to possess sufficiently smooth second-order derivatives. In many practical situations, such regularity assumptions are too restrictive.

The weak formulation is obtained by reformulating the problem in an integral sense. Let $v \in H_0^1(\Omega)$ be a test function. Multiplying the equation by v gives us

$$-\Delta u\, v = fv.$$

Integrating over Ω, we obtain:

$$\int_\Omega -\Delta u\, v\, dx = \int_\Omega fv\, dx.$$

We now apply Green's formula (integration by parts)

$$\int_\Omega -\Delta u\, v\, dx = \int_\Omega \nabla u \cdot \nabla v\, dx - \int_{\partial\Omega} \frac{\partial u}{\partial n} v\, ds.$$

Since $v = 0$ on $\partial\Omega$, the boundary term vanishes, and we obtain

$$\int_\Omega \nabla u \cdot \nabla v\, dx = \int_\Omega fv\, dx.$$

This leads to the weak formulation of the Poisson problem

$$\text{Find } u \in H_0^1(\Omega) \text{ such that } \int_\Omega \nabla u \cdot \nabla v\, dx = \int_\Omega fv\, dx, \quad \forall v \in H_0^1(\Omega).$$

In this formulation, the solution is no longer required to satisfy the differential equation pointwise. Instead, it must satisfy an integral identity for every test function in $H_0^1(\Omega)$. This weaker notion of solution allows functions with less regularity and forms the theoretical foundation of the finite element method.

The problem can be written in the following abstract variational form

$$\text{Find } u \in V \text{ such that } a(u, v) = F(v), \qquad \forall v \in V,$$

where

$$\begin{aligned} V &= H_0^1(\Omega), \\ a(u, v) &= \int_\Omega \nabla u \cdot \nabla v\, dx, \\ F(v) &= \int_\Omega fv\, dx. \end{aligned}$$

To study the well-posedness of this variational problem, we introduce the notion of coercivity.

Definition 7 (Coercive bilinear form)**.** *Let V be a Hilbert space and let $a : V \times V \to \mathbb{R}$ be a bilinear form. The form a is said to be* coercive *if there exists a constant $\alpha > 0$ such that*

$$a(v, v) \geq \alpha \|v\|_V^2, \qquad \forall v \in V.$$

We can now state the Lax-Milgram theorem[2].

Theorem 1 (Lax-Milgram)**.** *Let V be a Hilbert space, let $a : V \times V \to \mathbb{R}$ be a bilinear form that is continuous and coercive, and let $F \in V'$ be a continuous linear functional on V. Then the variational problem*

$$\text{Find } u \in V \text{ such that } a(u, v) = F(v), \qquad \forall v \in V,$$

admits a unique solution. Moreover, the solution satisfies the stability estimate

$$\|u\|_V \leq \frac{1}{\alpha} \|F\|_{V'}.$$

It is important to notice that in our case, The partial differential equation is not represented by coercive bilinear form. That is why we introduce the notion of inf-sup stability condition later.

4.4 Approximation of Variational Problems, the Galerkin Method

In the previous chapter, we introduced the weak formulation of partial differential equations and showed, using the Lax-Milgram theorem, that the variational problem admits a unique solution. However, this solution belongs to an infinite-dimensional Hilbert space and therefore cannot be computed directly in practice.

The main idea of the finite element method is to approximate the infinite-dimensional space V by a finite-dimensional subspace V_h. This allows the variational problem to be transformed into a finite system of equations that can be solved numerically. We consider the abstract variational problem

$$\text{Find } u \in V \text{ such that } a(u, v) = F(v), \qquad \forall v \in V.$$

Where V is a Hilbert space, $a(\cdot, \cdot)$ is a continuous coercive bilinear form, and F is a continuous linear functional on V. Under the assumptions of the Lax-Milgram theorem, this problem admits a unique solution.

To approximate this problem, we introduce a family of finite-dimensional spaces $V_h \subset V$, where the parameter $h > 0$ denotes the mesh size. The spaces V_h are assumed to satisfy two fundamental properties. First, the spaces are conforming, meaning that $V_h \subset V$. Second, they satisfy an approximability property

$$\forall w \in V, \qquad \lim_{h \to 0} \inf_{w_h \in V_h} \|w - w_h\|_V = 0.$$

This condition means that every function in V can be approximated arbitrarily well by functions belonging to V_h as the mesh size tends to zero.

The Galerkin approximation of the variational problem is then defined as follows

$$\text{Find } u_h \in V_h \text{ such that } a(u_h, v_h) = F(v_h), \qquad \forall v_h \in V_h.$$

Since V_h is finite dimensional, the approximate solution u_h can be computed numerically. Moreover, because $V_h \subset V$, the bilinear form $a(\cdot, \cdot)$ remains continuous and coercive on V_h. Therefore, the

discrete problem also satisfies the assumptions of the Lax-Milgram theorem and admits a unique solution.

The finite element method is obtained by choosing specific approximation spaces V_h, typically constructed from piecewise polynomial functions defined on a mesh of the domain. To transform the Galerkin formulation into a computable problem, we introduce a basis $\{\varphi_1, \ldots, \varphi_{N_h}\}$ of the space V_h, where $N_h = \dim(V_h)$. Any function $v_h \in V_h$ can then be written as

$$v_h = \sum_{j=1}^{N_h} v_j \varphi_j.$$

With $v_j \in \mathbb{R}$ In particular, the approximate solution u_h can be expanded as

$$u_h = \sum_{j=1}^{N_h} U_j \varphi_j.$$

Where the coefficients $U_j \in \mathbb{R}$ are unknown.

Substituting this expansion into the Galerkin formulation and testing against each basis function φ_i leads to

$$\sum_{j=1}^{N_h} U_j \, a(\varphi_j, \varphi_i) = F(\varphi_i), \qquad i = 1, \ldots, N_h.$$

Defining the matrix $A \in \mathbb{R}^{N_h \times N_h}$ and the vector $b \in \mathbb{R}^{N_h}$ by

$$A_{ij} = a(\varphi_j, \varphi_i), \qquad b_i = F(\varphi_i).$$

The discrete problem can be written in matrix form as

$$AU = b.$$

Where $U = (U_1, ..., U_{N_h})$, The matrix A is called the stiffness matrix. Solving the variational problem is therefore reduced to solving a finite-dimensional linear system.

4.5 Finite element space

A finite element space is a function space consisting of piecewise polynomial functions defined with respect to a decomposition of the domain Ω into non-overlapping subdomains called a mesh. These subdomains are typically simple geometric elements such as triangles in two dimensions or tetrahedra (or more general polyhedra) in three dimensions.

Finite element spaces can vary depending on several choices, including the degree of the polynomial basis functions, the type of elements used in the mesh, and the continuity requirements imposed across element interfaces.

In this chapter, we assume that the domain $\Omega \subset \mathbb{R}^d$ is a polygonal domain in two dimensions or a polyhedral domain in three dimensions. In our case, in two dimensions, we consider quadrilateral elements, whereas in three dimensions we employ polyhedral elements.

4.5.1 Mesh Discretization

A mesh of a domain Ω provides a discrete geometric decomposition of the domain into a finite number of simple subdomains [2]. This partition plays a central role in the finite element method, as it serves as the underlying structure on which finite-dimensional approximation spaces are constructed.

Definition 8 (Mesh)**.** *A polyhedral mesh $\mathcal{T}_h$ of the domain $\Omega \subset \mathbb{R}^d$ is a finite collection of polyhedral elements $K \subset \mathbb{R}^d$ such that*

- *the closure of the domain is exactly covered:*

$$\overline{\Omega} = \bigcup_{K \in \mathcal{T}_h} K,$$

- *the interiors of distinct elements are pairwise disjoint:*

$$\mathring{K}_i \cap \mathring{K}_j = \emptyset, \qquad \text{for } i \neq j.$$

The sets $K \in \mathcal{T}_h$ are called the elements *of the mesh.*

The most commonly used elements are triangles and quadrilaterals in 2D, and tetrahedra and hexahedra in 3D. In this work, we consider meshes made of cubic hexahedral elements. As hexahedral elements can, in principle, be arranged in many different ways to discretise a given domain, some additional structure is needed to make the mesh suitable for numerical methods such as the finite element method. Without such structure, it becomes difficult to guarantee consistency of the geometry, as well as the correct communication between neighbouring elements. For this reason, we introduce the notions of conformal and non-conformal meshes.

Definition 9. *A conformal mesh is a mesh such that, if $E = K_i \cap K_j \neq \emptyset$, then E is either a common vertex, a common edge, or (in 3D) a common face.*

Otherwise, if $\exists i \neq j$ such that $K_i \cap K_j \neq \emptyset$ is not a shared vertex, edge, or face of both elements, the mesh is called non-conformal.

Figure 1 below shows the differences between conforming and non-conforming meshes.

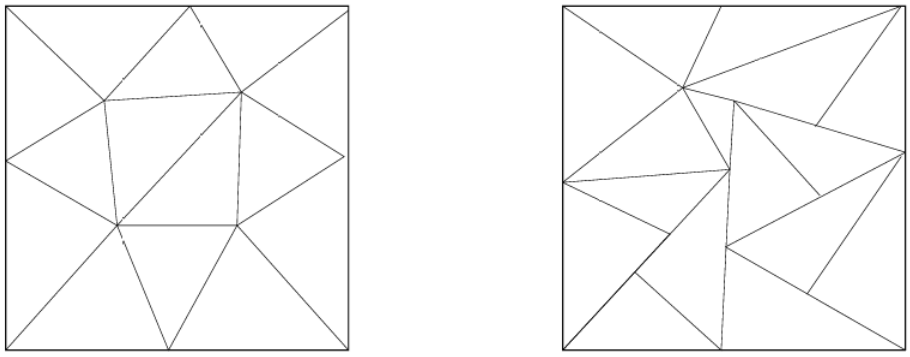


Figure 1: Example of a conformal mesh (left) and a non-conformal mesh (right) [2]

4.5.2 Mesh Parameters

Several geometric quantities are used to characterise the quality of a mesh. For an element $K \in \mathcal{T}_h$, its diameter is defined by

$$h_K = \text{diam}(K) = \max_{x,y \in K} |x - y|.$$

The quantity h_K measures the size of the element. The global mesh size is then defined as

$$h = \max_{K \in \mathcal{T}_h} h_K.$$

A second important quantity is the inner diameter ρ_K, defined as the diameter of the largest ball entirely contained in the element K.The ratio γ_k defined as

$$\gamma_K = \frac{h_K}{\rho_K},$$

is called the aspect ratio of the element and measures its distortion.

Definition 10. *A family of meshes $\{\mathcal{T}_h\}_{h \searrow 0}$ is said to be regular if there exists a constant $\gamma > 0$, independent of h, such that*

$$h_K \leq \gamma \rho_K, \qquad \forall K \in \mathcal{T}_h.$$

Regular meshes exclude excessively stretched elements and play an important role in the approximation properties of finite element spaces. As we want elements that globally have the same size we introduce the notion of quasi-uniform mesh.

Definition 11. *A family of regular meshes $\{\mathcal{T}_h\}_{h \searrow 0}$ is said to be quasi-uniform if $\exists 0 < \delta < 1 such that*

$$h_K \geq \delta h \quad \forall K \in \mathcal{T}_h.$$

With δ not dependent of h

4.5.3 Reference Element

Finite element basis functions are generally defined on a reference element and then mapped onto each element of the mesh. Since this work focuses on hexahedral meshes, the reference element is chosen as the unit cube

$$\widehat{K} = [0, 1]^3.$$

Each physical element $K \in \mathcal{T}_h$ is obtained from the reference element through a geometric mapping

$$F_K : \widehat{K} \to K.$$

We now state the notion of affine meshes which are meshes that possess elements that can be mapped to the reference element by an affine transformation.

Definition 12. *An affine mesh is a mesh for which each element K can be mapped onto the reference element $\hat{K}$ by an affine transformation,*

$$F_K(\widehat{x}) = B_K \widehat{x} + b_K.$$

Where $B_K \in \mathbb{R}^{3\times3}$ and $b_K \in \mathbb{R}^3$.

Figure 2 below shows the mapping from a reference unit cube to a real space hexahedron.

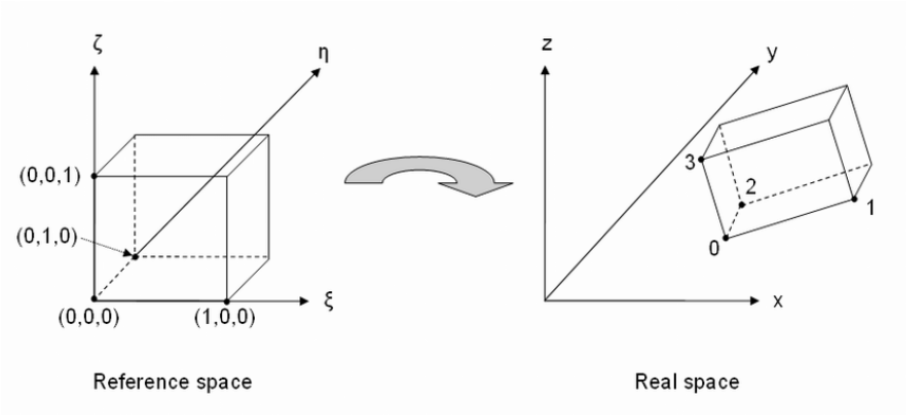


Figure 2: Mapping a reference unit cube to a real space hexahedron [10]

Using a reference element allows basis functions and numerical quadrature rules to be constructed only once and then reused on every element of the mesh.

4.5.4 Polynomial Approximation Spaces

Finite element spaces are constructed from polynomial functions defined on the reference element.

For hexahedral elements, tensor-product polynomial spaces are typically employed. Let p denote the polynomial degree. The space of tensor-product polynomials is

$$Q_p(\widehat{K}) = \left\{ v(\xi,\eta,\zeta) = \sum_{i,j,k=0}^{p} a_{ijk}\xi^i\eta^j\zeta^k \right\}.$$

The corresponding finite element space is obtained by requiring that the restriction of each function to an element K belongs to the image of $Q_p(\widehat{K})$ through the mapping F_K. One can check that the dimension of this family of spaces is

$$dim(Q_p) = (1+p)^3.$$

We can then define the space of continuous finite element on hexahedral mesh.

4.6 Continuous finite element space

Definition 13. *The space of continuous Finite Elements of degree p over a quadrilateral mesh $\mathcal{T}_h$ is defined for $p \geq 1$*

$$Y_h^p = \{v \in C^0(\Omega) | v_{|K} \circ F_K \in Q_p(\hat{K}) \forall K \in \mathcal{T}_h\}.$$

Let's set $p = 1$. In order to uniquely identify a linear polynomial in each hexahedral element we need to provide different values of the polynomial on eight different points of the hexahedron. These values are called the nodal degrees of freedom also called "dofs". For each hexahedron, the

clever choices of values are the one evaluated at each vertices of the hexahedron. Let N be the number of vertices of a hexahedron $K \in \mathcal{T}_h$ let $\{a_i\}_{i=1}^N$ the set of vertices of K we define a basis of Y_h^p in the following way:

$$\text{Basis of } Y_h^1: \quad \{\varphi_j \in Y_h^1, j = 1, ..., N\}: \quad \varphi_j(a_i) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} .$$

Then let $v_h \in Y_h^p$ we can express it as a linear combination of the basis which is

$$v_h(x) = \sum_{j=1}^{N} v_j \varphi_j(x), \quad \text{with } v_j = v_h(a_j) \in \mathbb{R}.$$

Finally, we use an interpolation operator that given a function $v \in C^0(\Omega)$ gives us its associated function in the finite element space v_h. The global interpolation operator I_h^1 is defined as

$$I_h^1 : C^0(\Omega) \to Y_h^1, \quad I_h^1(v) = v_h = \sum_{j=1}^{N} v(a_j)\varphi_j.$$

For $p = 2$. A natural choice of nodal degrees of freedom consists of: the 8 vertices, the 12 edge midpoints, the 6 face centers, and the interior point of the hexahedron.

Let $\{x_i\}_{i=1}^{27}$ denote these nodal points. The associated nodal basis $\{\varphi_i\}_{i=1}^{27} \subset Y_h^2$ is defined by the Kronecker property

$$\varphi_j(x_i) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Any function $v_h \in Y_h^2$ can be expressed locally on each element as follows:

$$v_h(x) = \sum_{j=1}^{27} v_j \varphi_j(x), \quad \text{with } v_j = v_h(x_j).$$

The interpolation operator $I_h^2 : C^0(\Omega) \to Y_h^2$ is defined by

$$I_h^2 v = \sum_{j=1}^{27} v(x_j)\, \varphi_j.$$

For $p > 2$ we keep this process. We can see in Figure 3 below the nodes selected for $p = 1, 2, 3$

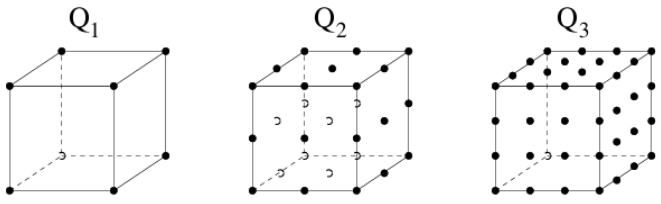


Figure 3: Example of point selected for Dofs for hexahedron and $p = 1, 2, 3$ [2]

In the discontinuous Galerkin framework, the finite element space is defined in $L^2(\Omega)$ instead of $C^0(\Omega)$

$$V_h^p = \left\{ v \in L^2(\Omega) \;\middle|\; v|_K \circ F_K \in Q_p(\widehat{K}),\ \forall K \in \mathcal{T}_h \right\}.$$

In this case, no continuity is enforced across element interfaces. Each element is treated independently, meaning that degrees of freedom are not shared between neighboring elements. As a consequence, nodal values are defined independently on each element: a given geometric point lying on an interface may take different values depending on the element in which it is evaluated.

5 Time integration

In this project, since the partial differential equations are time-dependent, we introduce a time discretization using Runge–Kutta methods. We first recall what Runge–Kutta methods are: Runge-Kutta methods form a family of numerical schemes used to approximate solutions of ordinary differential equations (ODEs) [1]. These methods work by evaluating the derivative (slope) at several points within a single time step, then combining these evaluations into a weighted average. This approach allows for higher accuracy without significantly increasing computational complexity.

Each specific Runge-Kutta method is defined by the number of slope evaluations it uses and how they are combined.
Consider the following ODE : given $f : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d$

$$\begin{cases} \dot{y}(t) = f(t, y(t)) & t \in [t_0, T], \\ y(t_0) = y_0. \end{cases}$$

We consider the following definition of Runge-Kutta methods.

Definition 14. *Let $N \in \mathbb{N}$ consider the discretization $\{t_0, t_0 + h, ..., T\}$ of $[t_0, T]$, where $h = \frac{T-t_0}{N}$, let $S \in \mathbb{N}^*$ and $b_i, a_{ij}, c_i \in \mathbb{R}$, $1 \leq i, j \leq S$. An S-stages Runge-Kutta method is given by :*

$$\begin{cases} k_i = f(t_n + c_i h, y_n + h \sum_{j=1}^{s} a_{ij} k_j), \\ y_{n+1} = y_n + h \sum_{i=1}^{s} b_i k_i. \end{cases}$$

Where $t_1 = t_0 + h$, $t_{n+1} = t_n + h$

Another way to represent an S-stages Runge-Kutta method is by considering the butcher table associated to it :

$$\begin{array}{c|cccc} c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\ c_2 & a_{21} & a_{22} & \cdots & a_{2s} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_s & a_{s1} & a_{s2} & \cdots & a_{ss} \\ \hline & b_1 & b_2 & \cdots & b_s \end{array}$$

We define the notion of explicit and implicit Runge-Kutta methods.

Definition 15. *We say that a Runge-Kutta method is explicit if $a_{ij} = 0$ for $i \geq j$. It is implicit otherwise.*

In order to understand the behavior of the error we define the notion of order within the Runge-Kutta methods.

Definition 16. *An S-stage Runge-Kutta method has order p if for a sufficiently smooth problem we have*

$$||y(t_n + h) - y_{n+1}|| \leq Ch^{p+1}.$$

In this project, we will only consider two different schemes: the order one implicit scheme :"implicit Euler" and the order two implicit scheme :"Crank-Nicolson", which correspond to the

two following Butcher tables:

1	1
	1

Implicit Euler

0	0	0
1	0.5	0.5
	0.5	0.5

Crank-Nicolson

6 Inf-sup stability theory

In order to understand existence, uniqueness, and stability of solutions in both the continuous and discrete settings, we introduce the notion of inf-sup stability following the framework of Brezzi and Fortin [4].

Let V be a Hilbert space endowed with the norm $\|\cdot\|_V$ and inner product $\langle\cdot,\cdot\rangle_V$. We consider a bilinear form $a(\cdot,\cdot)$ defined on $V \times V$, assumed to be continuous but not necessarily symmetric. This means that there exists a constant $C_a > 0$ such that

$$|a(u,v)| \leq C_a \|u\|_V \|v\|_V, \qquad \forall u, v \in V.$$

This bilinear form induces a bounded linear operator $A : V \to V'$ defined by the duality relation

$$\langle Au, v\rangle_{V',V} = a(u,v), \qquad \forall u, v \in V.$$

We now introduce a second Hilbert space Q, equipped with norm $\|\cdot\|_Q$ and inner product $\langle\cdot,\cdot\rangle_Q$, together with a continuous bilinear form $b(\cdot,\cdot)$ on $V \times Q$. There exists a constant $C_b > 0$ such that

$$|b(v,q)| \leq C_b \|v\|_V \|q\|_Q, \qquad \forall v \in V,\ \forall q \in Q.$$

This bilinear form defines a bounded operator $B : V \to Q'$ and its adjoint $B^t : Q \to V'$ through

$$\langle Bv, q\rangle_{Q',Q} = \langle v, B^t q\rangle_{V,V'} = b(v,q), \qquad \forall v \in V,\ \forall q \in Q.$$

Given $f \in V'$ and $g \in Q'$, we consider the saddle point problem: find $(u,p) \in V \times Q$ such that

$$\begin{cases} a(u,v) + b(v,p) = \langle f, v\rangle_{V'\times V}, & \forall v \in V, \\ b(u,q) = \langle g, q\rangle_{Q'\times Q}, & \forall q \in Q. \end{cases}$$

Equivalently, this system can be written in operator form as

$$\begin{cases} Au + B^t p = f \quad \text{in } V', \\ Bu = g \quad \text{in } Q'. \end{cases}$$

The objective is to identify conditions ensuring that this coupled system is well-posed, meaning that solutions exist and are unique in an appropriate sense.

Theorem 2. *[4] Assume that the range of $B : V \to Q'$ is closed in Q', i.e., there exists a constant $\beta > 0$ such that*

$$\sup_{v \in V} \frac{b(v,q)}{\|v\|_V} \geq \beta \, \|q\|_{Q'/\mathrm{Im}(B)}, \qquad \forall q \in Q.$$

With $\|v\|_{Q'/\mathrm{Im}(B)} = \inf_{v_0 \in Im(B)} \|v + v_0\|_{Q'}$.

Assume moreover that $a(\cdot,\cdot)$ *is coercive on* $\ker(B)$*, i.e., there exists* $\alpha > 0$ *such that*

$$a(u_0, u_0) \geq \alpha \|u_0\|_V^2, \qquad \forall u_0 \in \ker(B).$$

Then, for every $f \in V'$ *and every* $g \in \mathrm{Im}(B)$*, there exists a solution* $(u, p) \in V \times Q$ *to the previous system.*

Moreover, u *is uniquely determined, while* p *is unique up to an element of* $\ker(B^t)$*.*

We now consider the approximation of the previous problem: Let $V_h \subset V$ and $Q_h \subset Q$ be finite-dimensional subspaces associated with a discretization parameter h (e.g., a mesh size in finite element methods).

We seek $(u_h, p_h) \in V_h \times Q_h$ satisfying

$$\begin{cases} a(u_h, v_h) + b(v_h, p_h) = \langle f, v_h \rangle_{V' \times V}, & \forall v_h \in V_h, \\ b(u_h, q_h) = \langle g, q_h \rangle_{Q' \times Q}, & \forall q_h \in Q_h. \end{cases} \tag{1}$$

The stability of this discrete problem depends on a compatibility condition between V_h and Q_h, expressed through a discrete inf-sup type assumption. We introduce the following notation. For $g \in Q'$, we define

$$Z_h(g) = \{v_h \in V_h \mid b(v_h, q_h) = \langle g, q_h \rangle_{Q',Q},\ \forall q_h \in Q_h\}.$$

For $g = 0$, this set reduces to

$$Z_h(0) = \ker B_h.$$

We also define

$$\ker B_h^t = \{q_h \in Q_h \mid b(v_h, q_h) = 0,\ \forall v_h \in V_h\}.$$

A necessary condition for solvability of the discrete system is

$$Z_h(g) \neq \emptyset. \tag{2}$$

We assume that there exists $\alpha_h > 0$ such that

$$\inf_{u_h \in \ker B_h} \sup_{v_h \in \ker B_h} \frac{a(u_h, v_h)}{\|u_h\|_V \|v_h\|_V} \geq \alpha_h. \tag{3}$$

Proposition 1. *Assume that (2) and (3) holds. Then (1) admits at least one solution* (u_h, p_h)*. Moreover,* u_h *is uniquely determined in* V_h*, and* p_h *is uniquely determined in* $Q_h / \ker(B_h^t)$*.*

7 Different Problems Solved During the MFEM Learning Process

In this section, we validate the MFEM implementation through a series of benchmark problems with known analytical solutions. We start from a simple stationary mixed formulation and progressively increase the complexity by introducing time dependence and stronger coupling between variables. For each case, we present the discrete formulation and assess the numerical accuracy through convergence studies, in order to verify that the implementation reproduces the expected theoretical rates and remains stable across all configurations.

7.1 First Model Problem

As a first step toward understanding the implementation of mixed finite element methods in MFEM, we considered a simplified subsystem for which an analytical solution is available. This allowed us to verify the correctness of the discretization and become familiar with the assembly of the corresponding operators. We consider the following problem

$$\begin{cases} u - \nabla\phi = f, & \text{in } \Omega, \\ \nabla \cdot u = 0, & \text{in } \Omega, \\ \phi = 0, & \text{on } \partial\Omega. \end{cases}$$

Where $\Omega = [0,1]^2$ and

$$f = \begin{pmatrix} \sin(\pi x)\cos(\pi y) - \pi\cos(\pi x)\sin(\pi y) \\ -\cos(\pi x)\sin(\pi y) - \pi\sin(\pi x)\cos(\pi y) \end{pmatrix}.$$

The exact solutions are given by

$$\phi = \sin(\pi x)\sin(\pi y), \qquad u = \begin{pmatrix} \sin(\pi x)\cos(\pi y) \\ -\cos(\pi x)\sin(\pi y) \end{pmatrix}.$$

To derive the weak formulation, we multiply the first equation by a test function $v \in L^2(\Omega)$ and the second equation by a test function $w \in H^1(\Omega)$, and integrate over the domain. This yields

$$\begin{cases} \displaystyle\int_\Omega u \cdot v\,dx - \int_\Omega \nabla\phi \cdot v\,dx = \int_\Omega f \cdot v\,dx, & \forall v \in L^2(\Omega), \\ \displaystyle\int_\Omega (\nabla \cdot u)\, w\,dx = 0, & \forall w \in H^1(\Omega). \end{cases}$$

Choosing an L^2 finite element space for u and an H^1 finite element space for ϕ, the resulting discrete system can be written in block form as

$$\begin{pmatrix} M & -B \\ -B^T & 0 \end{pmatrix} \begin{pmatrix} u \\ \phi \end{pmatrix} = \begin{pmatrix} Mf \\ 0 \end{pmatrix}.$$

Where M denotes the finite element method mass matrix and B represents the discrete gradient operator. It is important to note that we use some abuse of notation: u and ϕ denote both the unknown functions and their corresponding vectors of degrees of freedom, while f represents the source term.

7.1.1 Numerical results

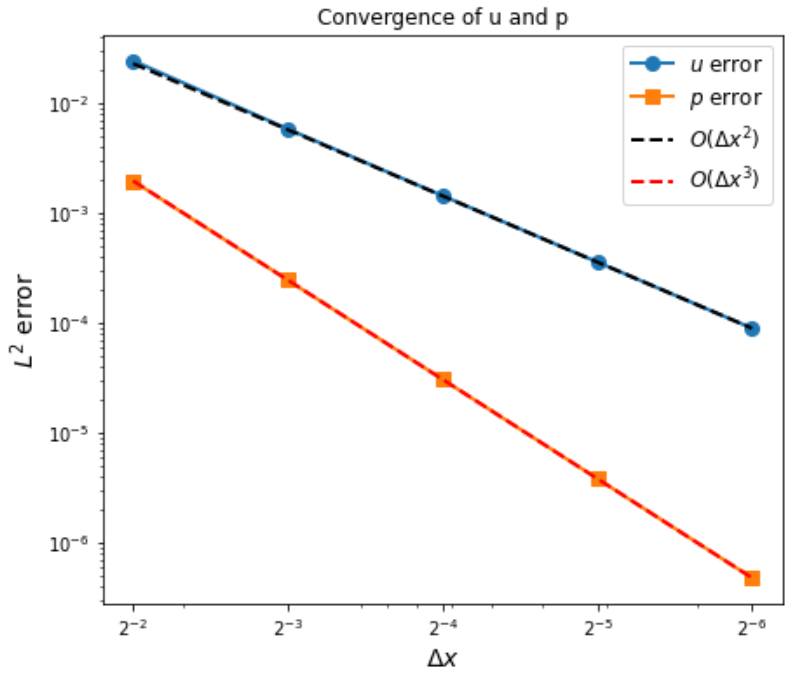


Figure 4: Convergence plot of the first model for the velocity u and the potential ϕ

Figure 4 clearly shows that both variables exhibit the expected asymptotic behaviour as the mesh is refined. The L^2-error for u decreases consistently with mesh refinement, following an approximately quadratic rate, which is consistent with the use of a P_1 finite element approximation. Similarly, the error for ϕ decreases even faster and aligns with an approximately cubic convergence rate, as expected for a higher-order P_2 approximation. The log-log convergence plot confirms these theoretical rates, with both error curves closely following the reference slopes $\mathcal{O}(\Delta x^2)$ for u and $\mathcal{O}(\Delta x^3)$ for p.

7.2 Second Model Problem

Having validated the spatial discretization on a stationary problem, we next consider a time-dependent extension in order to verify the implementation of temporal discretization schemes.

We consider

$$\begin{cases} u_t - \nabla\phi = f, & \text{in } \Omega, \\ \nabla \cdot u = 0, & \text{in } \Omega, \\ \phi = 0, & \text{on } \partial\Omega. \end{cases}$$

Where $\Omega = [0,1]^2$ and

$$f = \begin{pmatrix} e^t \sin(\pi x)\cos(\pi y) - \pi \cos(\pi x)\sin(\pi y) \\ -e^t \cos(\pi x)\sin(\pi y) - \pi \sin(\pi x)\cos(\pi y) \end{pmatrix}.$$

The exact solutions are given by

$$\phi = \sin(\pi x)\sin(\pi y), \qquad u = \begin{pmatrix} e^t \sin(\pi x)\cos(\pi y) \\ -e^t \cos(\pi x)\sin(\pi y) \end{pmatrix}.$$

7.2.1 Implicit euler scheme

For the time discretization, we employ the implicit Euler method. At time level t_{n+1}, the problem becomes

$$\begin{cases} \dfrac{u_{n+1} - u_n}{\Delta t} - \nabla \phi_{n+1} = f_{n+1}, \\ \nabla \cdot u_{n+1} = 0, \\ \phi_{n+1} = 0 \quad \text{on } \partial\Omega. \end{cases}$$

Multiplying the equations by test functions $v \in L^2(\Omega)$ and $w \in H^1(\Omega)$, respectively, and integrating over Ω, yields

$$\begin{cases} \dfrac{1}{\Delta t}\displaystyle\int_\Omega u_{n+1} \cdot v \, dx - \int_\Omega \nabla \phi_{n+1} \cdot v \, dx = \frac{1}{\Delta t}\int_\Omega u_n \cdot v \, dx + \int_\Omega f_{n+1} \cdot v \, dx & \forall v \in L^2(\Omega), \\ \displaystyle\int_\Omega (\nabla \cdot u_{n+1})\, w \, dx = 0 & \forall w \in H^1(\Omega). \end{cases}$$

Using the same finite element spaces as in the previous subsection, the corresponding discrete system can be written as

$$\begin{pmatrix} \dfrac{1}{\Delta t} M & -B \\ -B^T & 0 \end{pmatrix} \begin{pmatrix} u_{n+1} \\ \phi_{n+1} \end{pmatrix} = \begin{pmatrix} M f_{n+1} + \dfrac{1}{\Delta t} M u_n \\ 0 \end{pmatrix}.$$

7.2.2 Numerical results

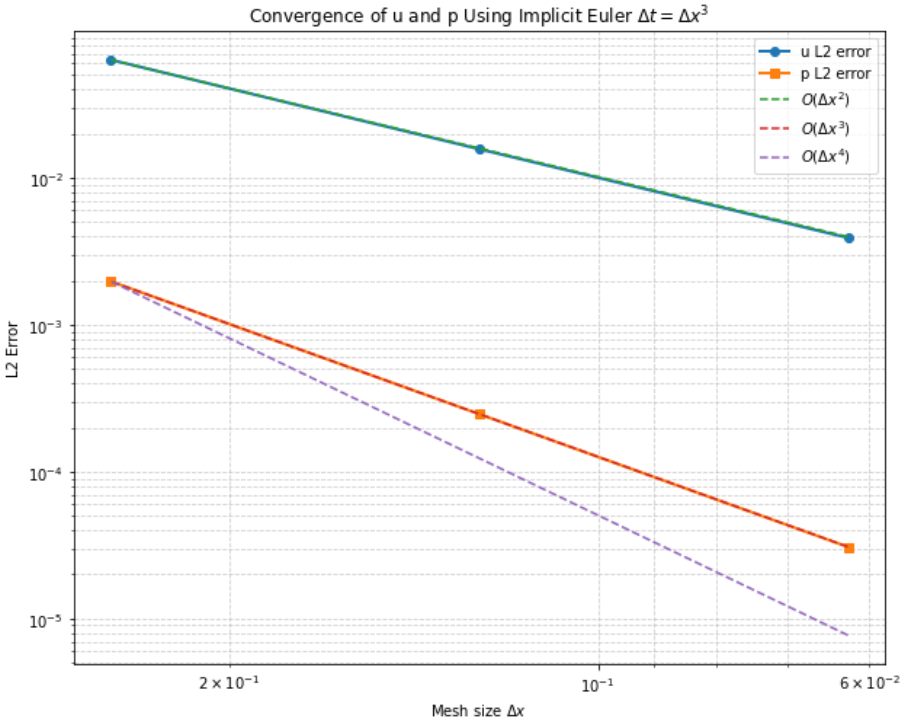


Figure 5: Convergence plot of the second model for the velocity u and the potential ϕ using implicit Euler with $\Delta t = \Delta x^3$

For the second model problem, we use a time step $\Delta t = \Delta x^3$.

Figure 5 shows that the error in u converges with order $\mathcal{O}(\Delta x^2)$, which is consistent with the use of a first-order L^2 finite element space for the velocity variable. In contrast, the error in ϕ converges with order $\mathcal{O}(\Delta x^3)$, reflecting the higher-order approximation associated with the second-order H^1 finite element space for the potential.

Overall, these results confirm that the implementation correctly reproduces the expected convergence rates of the mixed formulation and is consistent with the chosen finite element discretization spaces.

7.3 Third Model Problem

We now consider a coupled time-dependent system involving two velocity fields, u_i and u_e, and a scalar potential ϕ. This model introduces a stronger coupling between variables compared to the previous test cases and serves as a more advanced benchmark for validating the implementation of mixed and coupled finite element formulations in MFEM.

We consider the system

$$\begin{cases} \partial_t u_i + \nabla \phi = f_i, \\ \partial_t u_e - \nabla \phi = f_e, \\ \nabla \cdot (u_i - u_e) = 0, \\ \phi = 0 \quad \text{on } \partial\Omega. \end{cases}$$

With $\Omega = [0,1]^2$, and the two source terms

$$f_i = \begin{pmatrix} e^t \sin(\pi x) e^{\,1-\cos(\pi y)} \\ -e^{-t} \sin(\pi x)\cos(\pi y)\cos(\pi x) \end{pmatrix} + e^{t/10} \begin{pmatrix} \pi \cos(\pi x)\sin(\pi y)\big(1 - e^{1-(x^2+y^2)}\big) + 2x \sin(\pi x)\sin(\pi y)\, e^{1-(x^2+y^2)} \\ \pi \sin(\pi x)\cos(\pi y)\big(1 - e^{1-(x^2+y^2)}\big) + 2y \sin(\pi x)\sin(\pi y)\, e^{1-(x^2+y^2)} \end{pmatrix},$$

$$f_e = \begin{pmatrix} -\cos(\pi x)\cos(\pi y)\sin(t+5) + e^t \sin(\pi x) e^{1-\cos(\pi y)} \\ -\sin(\pi x)\sin(\pi y)\sin(t+5) - e^{-t}\sin(\pi x)\cos(\pi y)\cos(\pi x) \end{pmatrix}$$

$$-e^{t/10} \begin{pmatrix} \pi \cos(\pi x)\sin(\pi y)\big(1 - e^{1-(x^2+y^2)}\big) + 2x \sin(\pi x)\sin(\pi y) e^{1-(x^2+y^2)} \\ \pi \sin(\pi x)\cos(\pi y)\big(1 - e^{1-(x^2+y^2)}\big) + 2y \sin(\pi x)\sin(\pi y) e^{1-(x^2+y^2)} \end{pmatrix}.$$

The exact solutions are given by

$$u_i = \begin{pmatrix} e^t \sin(\pi x)\, e^{\,1-\cos(\pi y)} \\ e^{-t} \sin(\pi x)\ \cos(\pi y)\ \cos(\pi x) \end{pmatrix},$$

$$u_e = \begin{pmatrix} \cos(\pi x)\cos(\pi y)\cos(t+5) + e^t \sin(\pi x) e^{\,1-\cos(\pi y)} \\ \sin(\pi x)\sin(\pi y)\cos(t+5) + e^{-t}\sin(\pi x)\cos(\pi y)\cos(\pi x) \end{pmatrix},$$

$$\phi = e^{t/10} \sin(\pi x)\sin(\pi y)\Big(1 - e^{\,1-(x^2+y^2)}\Big).$$

7.3.1 Implicit Euler scheme

Applying the implicit Euler time discretization leads to the system at time step t_{n+1}:

$$\begin{cases} \dfrac{1}{\Delta t} u_i^{n+1} + \nabla \phi^{n+1} = \dfrac{1}{\Delta t} u_i^n + f_i^{n+1}, \\ \dfrac{1}{\Delta t} u_e^{n+1} - \nabla \phi^{n+1} = \dfrac{1}{\Delta t} u_e^n + f_e^{n+1}, \\ \nabla \cdot (u_i^{n+1} - u_e^{n+1}) = 0, \\ \phi = 0 \quad \text{on } \partial\Omega. \end{cases}$$

Multiplying by test functions and integrating over Ω yields the weak formulation

$$\begin{cases} \frac{1}{\Delta t}\int u_i^{n+1}\cdot v + \int \nabla\phi^{n+1}\cdot v = \frac{1}{\Delta t}\int u_i^{n}\cdot v + \int f_i^{n+1}\cdot v \quad \forall v \in L^2(\Omega), \\ \frac{1}{\Delta t}\int u_e^{n+1}\cdot v - \int \nabla\phi^{n+1}\cdot v = \frac{1}{\Delta t}\int u_e^{n}\cdot v + \int f_e^{n+1}\cdot v \quad \forall v \in L^2(\Omega), \\ \int \nabla\cdot(u_i^{n+1}-u_e^{n+1})\, w = 0 \quad \forall w \in H^1(\Omega). \end{cases}$$

This system can be written in block matrix form as

$$\begin{pmatrix} \frac{1}{\Delta t}M & 0 & B_i \\ 0 & \frac{1}{\Delta t}M & -B_e \\ B_i^T & -B_e^T & 0 \end{pmatrix} \begin{pmatrix} u_i^{n+1} \\ u_e^{n+1} \\ \phi^{n+1} \end{pmatrix} = \begin{pmatrix} Mf_i^{n+1} + \frac{1}{\Delta t}Mu_i^n \\ Mf_e^{n+1} + \frac{1}{\Delta t}Mu_e^n \\ 0 \end{pmatrix}.$$

7.3.2 Numerical results for Implicit Euler

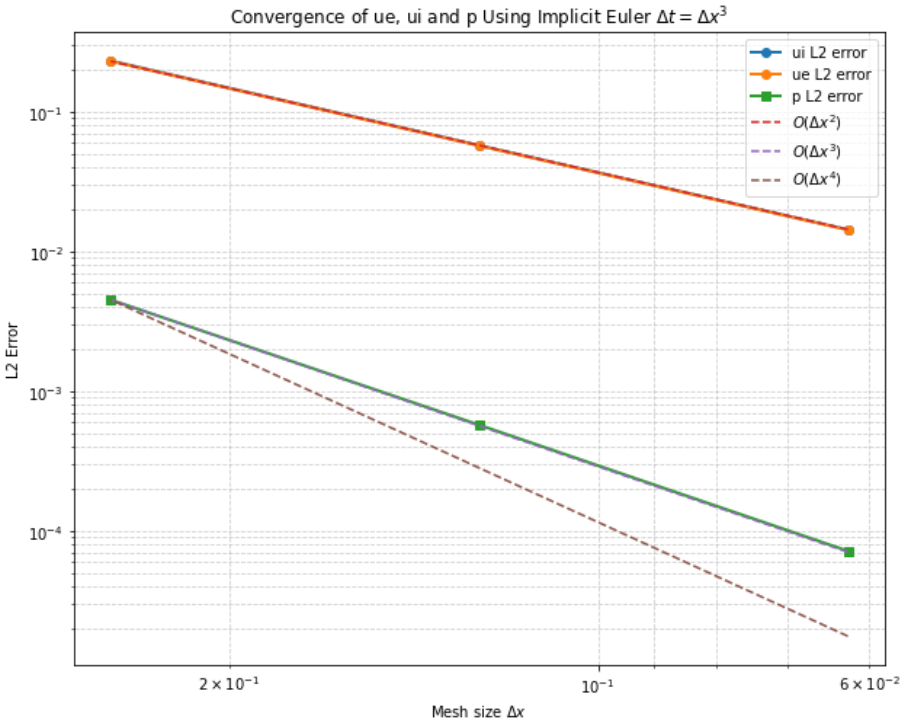


Figure 6: Convergence plot of the third model for the velocities u_i, u_e and the potential ϕ using implicit Euler with $\Delta t = \Delta x^3$

For the third model problem, we again choose a time step $\Delta t = \mathcal{O}(\Delta x^3)$.

Figure 6 shows that both u_i and u_e converge with order $\mathcal{O}(\Delta x^2)$, which is consistent with the use of a first-order L^2 finite element space for the velocity fields. In contrast, the error in ϕ converges with order $\mathcal{O}(\Delta x^3)$, reflecting the higher-order approximation associated with the second-order H^1 finite element space used for the potential.

The fact that all variables achieve their expected convergence rates confirms that the coupling between u_i, u_e, and ϕ does not degrade the accuracy of the discretization. Overall, the results validate both the spatial accuracy and the consistency of the coupled MFEM formulation.

7.3.3 Crank-Nicolson scheme

For improved temporal accuracy, we also consider the Crank-Nicolson discretization:

$$\begin{cases} \dfrac{1}{\Delta t}u_i^{n+1} + \nabla\left(\dfrac{\phi^{n+1}+\phi^n}{2}\right) = \dfrac{1}{\Delta t}u_i^n + \dfrac{f_i^{n+1}+f_i^n}{2}, \\ \dfrac{1}{\Delta t}u_e^{n+1} - \nabla\left(\dfrac{\phi^{n+1}+\phi^n}{2}\right) = \dfrac{1}{\Delta t}u_e^n + \dfrac{f_e^{n+1}+f_e^n}{2}, \\ \nabla\cdot(u_i^{n+1}-u_e^{n+1}) = 0, \\ \\ \phi = 0 \quad \text{on } \partial\Omega. \end{cases}$$

Proceeding as before, the weak formulation reads

$$\begin{cases} \dfrac{1}{\Delta t}\displaystyle\int u_i^{n+1}\cdot v + \int \nabla\left(\dfrac{\phi^{n+1}+\phi^n}{2}\right)\cdot v = \dfrac{1}{\Delta t}\int u_i^n\cdot v + \int \dfrac{f_i^{n+1}+f_i^n}{2}\cdot v \quad \forall v\in L^2(\Omega), \\ \dfrac{1}{\Delta t}\displaystyle\int u_e^{n+1}\cdot v - \int \nabla\left(\dfrac{\phi^{n+1}+\phi^n}{2}\right)\cdot v = \dfrac{1}{\Delta t}\int u_e^n\cdot v + \int \dfrac{f_e^{n+1}+f_e^n}{2}\cdot v \quad \forall v\in L^2(\Omega), \\ \displaystyle\int \nabla\cdot(u_i^{n+1}-u_e^{n+1})\,w = 0 \quad \forall w\in H^1(\Omega). \end{cases}$$

The corresponding linear system is

$$\begin{pmatrix} \frac{1}{\Delta t}M & 0 & \frac{1}{2}B_i \\ 0 & \frac{1}{\Delta t}M & -\frac{1}{2}B_e \\ \frac{1}{2}B_i^T & -\frac{1}{2}B_e^T & 0 \end{pmatrix} \begin{pmatrix} u_i^{n+1} \\ u_e^{n+1} \\ \phi^{n+1} \end{pmatrix} = \begin{pmatrix} M\left(\frac{f_i^{n+1}+f_i^n}{2}\right) + \frac{1}{\Delta t}Mu_i^n - \frac{1}{2}B_i\phi^n \\ M\left(\frac{f_e^{n+1}+f_e^n}{2}\right) + \frac{1}{\Delta t}Mu_e^n + \frac{1}{2}B_e\phi^n \\ 0 \end{pmatrix}.$$

7.3.4 Numerical results for Crank-Nicolson

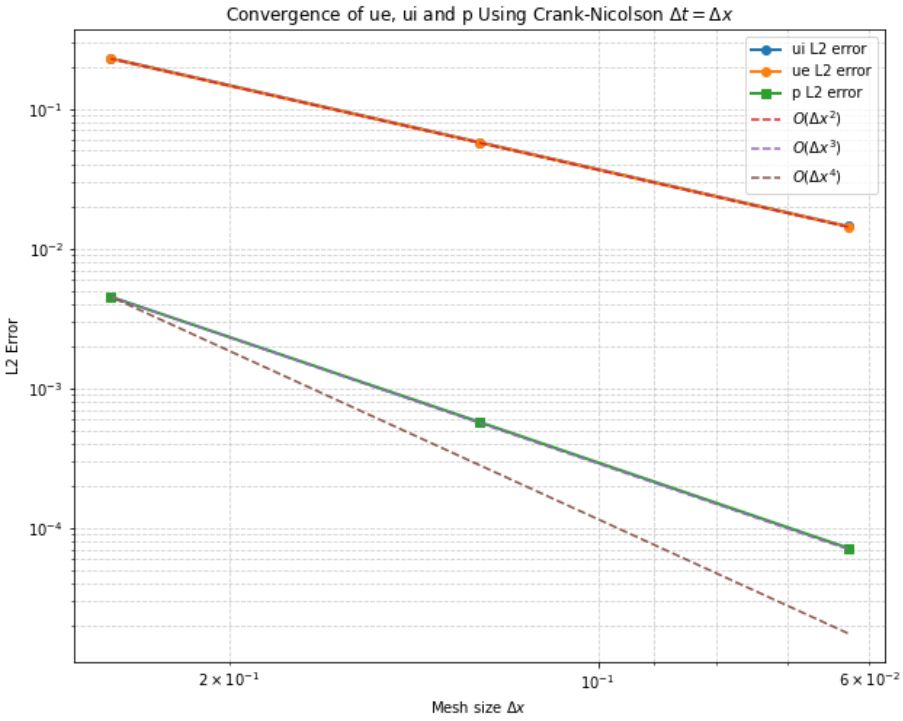


Figure 7: Convergence plot of the third model for the velocities u_i, u_e and the potential ϕ using Crank-Nicolson with $\Delta t = \Delta x$

For the Crank-Nicolson scheme, we consider a time step $\Delta t = \mathcal{O}(\Delta x)$ in order to study the coupled effect of spatial and temporal discretization at comparable scales.

Figure 7 shows that both u_i and u_e converge with order $\mathcal{O}(\Delta x^2)$, while the error in ϕ converges with order $\mathcal{O}(\Delta x^3)$. These rates are consistent with the spatial finite element discretization used for each variable and confirm that the Crank-Nicolson time integration does not degrade the spatial accuracy of the scheme.

In particular, the preservation of second-order convergence for u_i and u_e, together with the higher-order behavior observed for ϕ, indicates that the fully coupled discretization remains stable and accurate under this time integration strategy.

7.4 Fourth Model Problem

We finally consider an extension of the previous coupled system by including a magnetic-type advection term. This model introduces additional skew-symmetric coupling through cross-product operators and provides a more complete test of the MFEM implementation in a three-dimensional setting.

We consider the system

$$\begin{cases} \partial_t u_i + \nabla\phi - u_i \times B = f_i, \\ \partial_t u_e - \nabla\phi + u_e \times B = f_e, \\ \nabla \cdot (u_i - u_e) = 0, \\ \phi = 0 \quad \text{on } \partial\Omega_x, \partial\Omega_z. \end{cases}$$

With $\Omega = [0,1]^3$ with y-periodic boundary and $B = (0,1,0)$ and the source terms:

$$f_i = \begin{pmatrix} e^t\cos(\pi x) \\ e^t\cos(2\pi y) \\ e^t\sin(\pi z) \end{pmatrix} + \begin{pmatrix} \pi\cos(t)\cos(2\pi y)\cos(\pi x)\sin(\pi z) \\ -2\pi\cos(t)\sin(2\pi y)\sin(\pi x)\sin(\pi z) \\ \pi\cos(t)\cos(2\pi y)\sin(\pi x)\cos(\pi z) \end{pmatrix} - \begin{pmatrix} -e^t\sin(\pi z) \\ 0 \\ e^t\cos(\pi x) \end{pmatrix},$$

$$f_e = \begin{pmatrix} e^t\cos(\pi x) \\ e^t\cos(2\pi y) \\ e^t\sin(\pi z) \end{pmatrix} - \begin{pmatrix} \pi\cos(t)\cos(2\pi y)\cos(\pi x)\sin(\pi z) \\ -2\pi\cos(t)\sin(2\pi y)\sin(\pi x)\sin(\pi z) \\ \pi\cos(t)\cos(2\pi y)\sin(\pi x)\cos(\pi z) \end{pmatrix} + \begin{pmatrix} -e^t\sin(\pi z) \\ 0 \\ e^t\cos(\pi x) - \sin(2\pi y) \end{pmatrix}.$$

The corresponding exact solutions are given by

$$u_i = \begin{pmatrix} e^t\cos(\pi x) \\ e^t\cos(2\pi y) \\ e^t\sin(\pi z) \end{pmatrix}, \qquad u_e = \begin{pmatrix} e^t\cos(\pi x) - \sin(2\pi y) \\ e^t\cos(2\pi y) \\ e^t\sin(\pi z) \end{pmatrix},$$

$$\phi = \cos(t)\cos(2\pi y)\sin(\pi x)\sin(\pi z).$$

7.4.1 Crank-Nicolson scheme

Using the Crank-Nicolson time discretization, we obtain the following system at time level t_{n+1}:

$$\begin{cases} \dfrac{u_i^{n+1} - u_i^n}{\Delta t} + \nabla\left(\dfrac{\phi^{n+1} + \phi^n}{2}\right) - \left(\dfrac{u_i^{n+1} + u_i^n}{2}\right) \times B = \dfrac{f_i^{n+1} + f_i^n}{2}, \\ \dfrac{u_e^{n+1} - u_e^n}{\Delta t} - \nabla\left(\dfrac{\phi^{n+1} + \phi^n}{2}\right) + \left(\dfrac{u_e^{n+1} + u_e^n}{2}\right) \times B = \dfrac{f_e^{n+1} + f_e^n}{2}, \\ \nabla \cdot (u_i^{n+1} - u_e^{n+1}) = 0, \\ \phi^{n+1} = 0 \quad \text{on } \partial\Omega_x, \partial\Omega_z. \end{cases}$$

Multiplying by test functions and integrating over Ω, we obtain the weak formulation

$$\begin{cases} \displaystyle\int \frac{u_i^{n+1} - u_i^n}{\Delta t} \cdot v + \int \nabla\left(\frac{\phi^{n+1} + \phi^n}{2}\right) \cdot v - \int \left(\frac{u_i^{n+1} + u_i^n}{2} \times B\right) \cdot v = \int \frac{f_i^{n+1} + f_i^n}{2} \cdot v, \quad \forall v \in L^2(\Omega), \\ \displaystyle\int \frac{u_e^{n+1} - u_e^n}{\Delta t} \cdot v - \int \nabla\left(\frac{\phi^{n+1} + \phi^n}{2}\right) \cdot v + \int \left(\frac{u_e^{n+1} + u_e^n}{2} \times B\right) \cdot v = \int \frac{f_e^{n+1} + f_e^n}{2} \cdot v, \quad \forall v \in L^2(\Omega), \\ \displaystyle\int \nabla \cdot (u_i^{n+1} - u_e^{n+1})\, w = 0, \quad \forall w \in H^1(\Omega). \end{cases}$$

This can be written in block matrix form as

$$\begin{pmatrix} \frac{1}{\Delta t}M - \frac{1}{2}C_i & 0 & \frac{1}{2}b_i \\ 0 & \frac{1}{\Delta t}M + \frac{1}{2}C_e & -\frac{1}{2}b_e \\ \frac{1}{2}b_i^T & -\frac{1}{2}b_e^T & 0 \end{pmatrix} \begin{pmatrix} u_i^{n+1} \\ u_e^{n+1} \\ \phi^{n+1} \end{pmatrix} = \begin{pmatrix} M\left(\frac{f_i^{n+1}+f_i^n}{2}\right) + \left(\frac{1}{\Delta t}M + \frac{1}{2}C_i\right)u_i^n - \frac{1}{2}b_i\phi^n \\ M\left(\frac{f_e^{n+1}+f_e^n}{2}\right) + \left(\frac{1}{\Delta t}M - \frac{1}{2}C_e\right)u_e^n + \frac{1}{2}b_e\phi^n \\ 0 \end{pmatrix}.$$

7.4.2 Numerical results for Crank-Nicolson

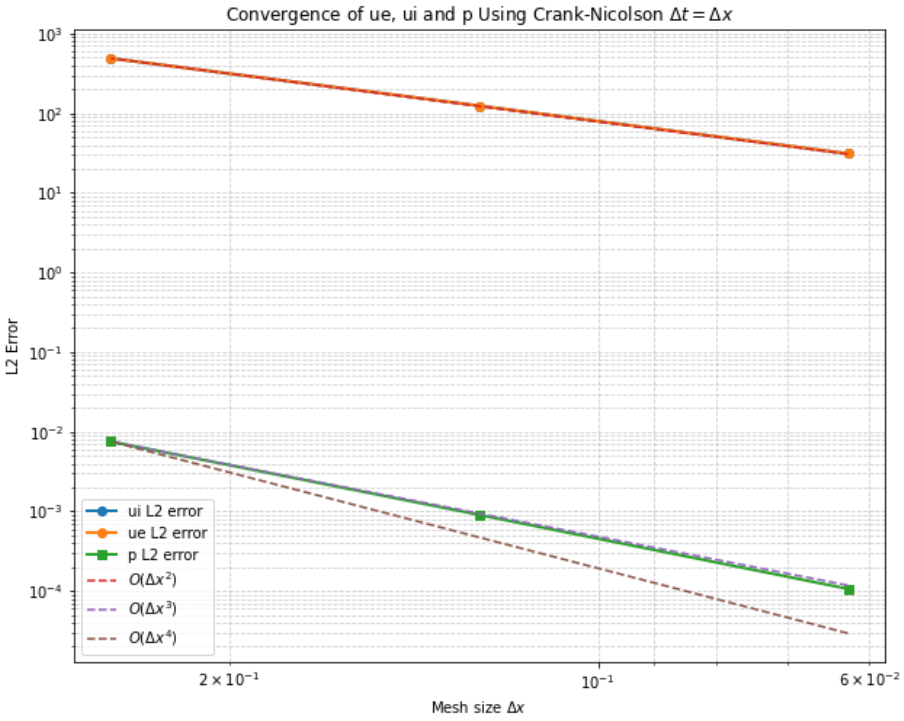


Figure 8: Convergence plot of the fourth model for the velocities u_i, u_e and the potential ϕ using Crank-Nicolson with $\Delta t = \Delta x$

For the Crank-Nicolson scheme applied to the fourth model problem, we choose a time step $\Delta t = \mathcal{O}(\Delta x)$ in order to study the coupled effect of spatial and temporal discretization at comparable scales.

Figure 8 shows that both u_i and u_e converge with order $\mathcal{O}(\Delta x^2)$, while the error in ϕ converges with order $\mathcal{O}(\Delta x^3)$. These rates are consistent with the spatial finite element discretization used for each variable and confirm that the inclusion of the magnetic-type advection term $u \times B$ does not affect the expected convergence behaviour.

Overall, the results indicate that the Crank-Nicolson scheme preserves the spatial accuracy of the coupled MFEM formulation even in the presence of additional terms, and that the implementation remains stable and consistent in the three-dimensional setting.

7.5 Conclusion of the numerical validation

In this chapter, we have progressively validated the MFEM implementation through a hierarchy of model problems of increasing complexity. Starting from a stationary mixed formulation, we first verified the correctness of the spatial discretization and the saddle-point structure of the resulting linear system. We then extended the analysis to time-dependent problems, where both implicit Euler and Crank-Nicolson schemes were tested.

Across all cases, the numerical experiments consistently demonstrate optimal spatial convergence rates: first-order accuracy for the velocity variables in the chosen L^2-based approximation, and higher-order convergence for the scalar potential associated with its H^1 discretization.

Finally, the more advanced coupled models, including multi-velocity interactions and magnetic-type advection terms, confirm that the implementation remains stable and accurate.

Overall, these numerical results validate the correctness, stability, and accuracy of the implemented MFEM framework across a range of representative test cases.

We now turn our attention to the more challenging regime involving a small parameter $\varepsilon \ll 1$. In this setting, the resulting linear systems become increasingly ill-conditioned as ε decreases, which significantly affects both the stability of the numerical solution and the efficiency of iterative solvers. This regime is particularly delicate from an implementation point of view, as it introduces stiffness and near-singular behavior in the discrete system matrices. The next chapters are therefore dedicated to the analysis and numerical treatment of this singularly perturbed problem.

8 Lagrange Multipliers theory

Lagrange multipliers [9] are a simple but very powerful tool used to "link" different quantities within a problem. Instead of reformulating the problem to eliminate constraints, an additional variable is introduced in order to enforce them.

A typical situation occurs when two different materials are considered. Each material can be described independently, but at their interface certain conditions must still be satisfied, such as continuity of a field or balance between quantities on both sides. Lagrange multipliers allow these conditions to be imposed in a natural way. In this context, they act as interface quantities that enforce the coupling between the two materials, and they often correspond to physically meaningful transfer terms across the interface.

They are also widely used in numerical methods. In particular, they make it possible to connect different computational domains even when their discretizations (meshes) are not compatible. This allows one to use a refined mesh only where it is necessary, while keeping a coarser mesh elsewhere, while still ensuring a consistent transfer of information between the two regions.

In summary, Lagrange multipliers provide a flexible way to impose constraints and to couple different parts of a model, both at the physical level between materials and at the numerical level between computational domains.

8.1 Constrained Minimization and Lagrange Multipliers

We define the Lagrange multipliers in the following context [9] :

Let H and $\mathcal{M}$ be two Hilbert spaces. Let $a(\cdot,\cdot) : H \times H \to \mathbb{R}$ be a continuous symmetric bilinear form and $b(\cdot,\cdot) : H \times \mathcal{M} \to \mathbb{R}$ be a continuous bilinear form. Furthermore, let $f \in H'$ and $g \in \mathcal{M}'$ be continuous linear functionals. The duality pairings between H and H', and between $\mathcal{M}$ and $\mathcal{M}'$, are denoted by $\langle\cdot,\cdot\rangle_H$ and $\langle\cdot,\cdot\rangle_{\mathcal{M}}$, respectively.

Consider the problem of finding $u \in H$ such that

$$a(u,v) = \langle f, v\rangle_H, \qquad \forall v \in H,$$

while satisfying the additional constraint

$$b(u,\mu) = \langle g, \mu\rangle_{\mathcal{M}}, \qquad \forall \mu \in \mathcal{M}.$$

This problem can be reformulated as a constrained minimization problem. Defining the functional

$$J(v) = \frac{1}{2}a(v,v) - \langle f, v\rangle_H,$$

the objective is to determine

$$u = \arg\min_{v\in H} J(v),$$

subject to

$$b(v,\mu) = \langle g, \mu\rangle_{\mathcal{M}}, \qquad \forall \mu \in \mathcal{M}.$$

To enforce the constraint, a Lagrange multiplier $\lambda \in \mathcal{M}$ is introduced through the Lagrangian functional

$$L(v, \lambda) = J(v) + b(v, \lambda) - \langle g, \lambda \rangle_{\mathcal{M}}.$$

The solution is obtained by seeking a saddle point $(u, \lambda) \in H \times \mathcal{M}$ of L. The stationarity condition with respect to v yields

$$a(u, v) + b(v, \lambda) = \langle f, v \rangle_H, \qquad \forall v \in H,$$

while the variation with respect to λ recovers the constraint

$$b(u, \mu) = \langle g, \mu \rangle_{\mathcal{M}}, \qquad \forall \mu \in \mathcal{M}.$$

The resulting mixed formulation therefore reads: find $(u, \lambda) \in H \times \mathcal{M}$ such that

$$\begin{aligned} a(u, v) + b(v, \lambda) &= \langle f, v \rangle_H \quad \forall v \in H, \\ b(u, \mu) &= \langle g, \mu \rangle_{\mathcal{M}} \quad \forall \mu \in \mathcal{M}. \end{aligned}$$

Finally, introducing the bilinear form

$$A\big((u, \lambda), (v, \mu)\big) = a(u, v) + b(v, \lambda) + b(u, \mu),$$

and the linear functional

$$L(v, \mu) = \langle f, v \rangle_H + \langle g, \mu \rangle_{\mathcal{M}},$$

the mixed problem can be expressed in the compact form: find $U = (u, \lambda) \in H \times \mathcal{M}$ such that

$$A(U, V) = L(V), \qquad \forall V = (v, \mu) \in H \times \mathcal{M}.$$

This formulation highlights the connection with variational problems of the Lax-Milgram type posed on the product space $H \times \mathcal{M}$.

9 Adapted method

We consider the system

$$\begin{cases} \partial_t u_i - \Big(-\nabla\phi + u_i \times B\Big) = f_i, \\ \epsilon\partial_t u_e + \Big(-\nabla\phi + u_e \times B\Big) = \epsilon f_e, \\ \nabla \cdot (u_i - u_e) = 0, \\ \phi = 0 \quad \text{on } \partial\Omega_x, \partial\Omega_z. \end{cases} \tag{4}$$

With $\Omega = [0,1]^3$ with y-periodic boundary and $B = (0,1,0)$.

The main difficulty with this system is that it becomes ill-posed in the singular limit $\epsilon \to 0$.

Proposition 2. *The system (4) becomes ill-posed when $\epsilon \to 0$*

Proof. We consider the y-component of the second equation of the system, we obtain :

$$\epsilon\partial_t u_{ey} - \partial_y\phi = \epsilon f_{ey},$$

which formally reduces, as $\epsilon \to 0$, to

$$\partial_y\phi = 0.$$

In particular, in the limiting system there is no equation involving the unknown u_{ey}. Let u_e be a solution of the limit system, and let $g = g(x,z,t)$ be any smooth function independent of y. We define

$$\widetilde{u}_e = u_e + (0, g, 0).$$

Then we have

$$\nabla \cdot (u_i - \widetilde{u}_e) = \nabla \cdot (u_i - u_e) - \partial_y g = \nabla \cdot (u_i - u_e),$$

since $\partial_y g = 0$, the divergence constraint is preserved. Moreover, the remaining equations of the limiting system are unaffected because they do not involve u_{ey}. Therefore, $\widetilde{u}_e$ is also a solution of the limiting system whenever u_e is a solution.

Since g can be chosen arbitrarily, the limiting system admits an infinite-dimensional family of solutions. Therefore, uniqueness is lost in the limit $\epsilon \to 0$, which shows that the limiting problem is ill-posed. □

This degeneracy can also be interpreted as the fact that the limiting system does not determine the y-component of u_e, which effectively leads to a non-trivial kernel in the set of admissible solutions. This problem has a direct numerical consequence: as ϵ becomes smaller and smaller, solving the system becomes significantly more computationally expensive, and in the limit case it becomes essentially intractable.

To bypass this difficulty, we use a splitting method introduced by Pierre Degond [7][6], which consists in decomposing the y-component of u_e into the sum of two parts belonging two orthogonal spaces.

9.1 Splitting method

In the paper "An asymptotic preserving scheme for strongly anisotropic elliptic problems" [6] Pierre Degond introduces a method to solve the system :

$$\begin{cases} -\nabla \cdot (\mathbf{A}\nabla\phi) = f & \text{in } \Omega, \\ \dfrac{\partial \phi}{\partial z} = 0 & \text{on } \Omega_x \times \partial\Omega_z, \\ \phi = 0 & \text{on } \partial\Omega_x \times \Omega_z. \end{cases}$$

With the matrix $\mathbf{A}$ defined in the following way :

$$\mathbf{A} = \begin{pmatrix} A_\perp & 0 \\ 0 & \frac{1}{\varepsilon} A_z \end{pmatrix},$$

where $A_\perp(x,z)$ and $A_z(x,z)$ are given functions with comparable orders of magnitude. The source term $f(x,z)$ is given and the parameter ε is small compared to both $A_\perp$ and A_z.

In this system, setting $\varepsilon = 0$ in the equation leads to a degenerate limit problem, which is no longer well-posed and admits an infinite family of solutions, namely all functions that are constant in the z direction.

Although the two systems come from different physical and mathematical settings, they share a common feature in the singular limit $\varepsilon \to 0$. In both cases, the presence of the small parameter leads to a loss of control on part of the solution, which in turn causes a loss of uniqueness in the limiting problem. As a result, the limit system becomes ill-posed and admits an infinite family of solutions. In order to avoid the ill-posed system, we follow the idea introduced in [6], which consists in writing the solution as a sum of two parts: one part lies in the space of functions that are constant along the direction that causes the loss of well-posedness, and the other part lies in its orthogonal space. In our case, one could decompose the full vector u_e, but since the loss of constraint only affects the y-component, we restrict the splitting to this component, which is computationally less expensive. Indeed, a full decomposition would increase the number of unknowns from 3 basic component unknowns to 6, whereas splitting only the y-component introduces only 4 unknowns.

Let V be the finite element space corresponding to $L^2(\Omega)$ and V_i be its one dimension part. We introduce the two spaces

$$G = \{v \in V_i \mid \partial_y v = 0\}, \qquad A = G^\perp = \left\{ v \in V_i \mid \int_0^{L_y} v \, dy = 0 \right\},$$

and we write u_{ey} as a direct sum:

$$u_{ey} = \overline{u_{ey}} + u'_{ey},$$

with $\overline{u_{ey}} \in G$ and $u'_{ey} \in A$.

This splitting is justified as follows. Let $u_y(x,y,z) \in V_i$. We define its average in the y-direction by

$$\overline{u_y}(x,z) = \frac{1}{L_y} \int_0^{L_y} u_y(x,y,z)\, dy,$$

with L_y denoting the length of the domain in the y-direction. We define the fluctuation component by

$$u'_y(x,y,z) = u_y(x,y,z) - \overline{u_y}(x,z).$$

By construction, we clearly have $\overline{u_y}(x,z) \in G$, while $u_y'(x,y,z) \in A$. Indeed,

$$u_y(x,y,z) = u_y'(x,y,z) + \overline{u_y}(x,z).$$

Taking the average in the y-direction on both sides yields

$$\overline{u_y}(x,z) = \overline{u_y'}(x,z) + \overline{u_y}(x,z),$$

which implies

$$\overline{u_y'}(x,z) = 0,$$

and therefore

$$\int_0^{L_y} u_y'(x,y,z)\, dy = 0.$$

This splitting works because it isolates the part of u_{ey} that is responsible for the loss of uniqueness in the limit system. Indeed, the degeneracy appears exactly in the y-independent component, which corresponds to the space G. This is precisely the part represented by $\overline{u_{ey}}$. In the limit $\varepsilon \to 0$, the equations do not provide any evolution or constraint on this component, so it is not determined by the system.

By identifying this term explicitly through the decomposition $u_{ey} = \overline{u_{ey}} + u_{ey}'$, we can directly fix $\overline{u_{ey}}$ using the structure of the equations, while u_{ey}' remains the only truly unknown part. In this way, the ambiguity coming from the kernel is removed by prescribing the evolution of $\overline{u_{ey}}$, which restores uniqueness of the solution. The condition defining A (zero mean in y) ensures that u_{ey}' contains no contribution from this undetermined mode.

9.2 New system

We now rewrite the system using the splitting $u_{ey} = \overline{u_{ey}} + u_{ey}'$, where $\overline{u_{ey}} \in G$ and $u_{ey}' \in A$. The main idea is that the y-independent part $\overline{u_{ey}}$, which lies in the kernel of the limiting operator, is no longer implicitly determined by the coupled system. To handle this degeneracy, we project the equations onto the spaces G and A, which leads to separate evolution equations for $\overline{u_{ey}}$ and u_{ey}'.

In this formulation, the component u_{ey}' evolves in the space A, where the zero-mean condition removes the kernel effect and restores uniqueness. The component $\overline{u_{ey}}$ is obtained through the projection onto G, which corresponds to the y-averaged part of the original y-component equation.

This splitting removes the non-uniqueness associated with the kernel and rewrites the system in a form where all components are explicitly determined. As a consequence, the reformulated system remains well-posed uniformly in ε, and in particular it is still well-posed in the limit $\varepsilon \to 0$.

Using the splitting method $u_{ey} = \overline{u_{ey}} + u'_{ey}$, we obtain the following system:

$$\begin{cases} \partial_t u_i - \Big(-\nabla\phi + u_i \times B\Big) = f_i, \\ \epsilon\partial_t u_{ex} + \Big(-\partial_x\phi - u_{ez}\Big) = \epsilon f_{ex}, \\ \epsilon\partial_t(u'_{ey} + \overline{u_{ey}}) + \Big(-\partial_y\phi\Big) = \epsilon f_{ey}, \\ \epsilon\partial_t u_{ez} + \Big(-\partial_z\phi + u_{ex}\Big) = \epsilon f_{ez}, \\ \nabla \cdot (u_i - u_e) = 0, \\ \phi = 0 \quad \text{on } \partial\Omega_x \cup \partial\Omega_z. \end{cases}$$

We then multiply each equation by test functions in the corresponding spaces and integrate in space, which yields the weak formulation:

$$\begin{cases} \displaystyle\int \partial_x(u_{ix}^{n+1} - u_{ex}^{n+1})v + \int \partial_z(u_{iz}^{n+1} - u_{ez}^{n+1})v + \int \partial_y(u_{iy}^{n+1} - u_{ey}'^{n+1} - \overline{u_{ey}}^{n+1})v = 0 \quad \forall v \in H^1(\Omega), \\ \displaystyle\frac{1}{\Delta t}\int u_{ex}^{n+1}v - \frac{1}{\epsilon}\int \partial_x\Phi^{n+1}v - \frac{1}{\epsilon}\int u_{ez}^{n+1}v = \frac{1}{\Delta t}\int u_{ex}^{n}v + \int f_{ex}^{n+1}v \quad \forall v \in V_i, \\ \displaystyle\frac{1}{\Delta t}\int u_{ey}'^{n+1}v - \frac{1}{\epsilon}\int \partial_y\Phi^{n+1}v = \frac{1}{\Delta t}\int u_{ey}'^{n}v + \int f_{ey}^{n+1}v \quad \forall v \in A, \\ \displaystyle\frac{1}{\Delta t}\int \overline{u_{ey}}^{n+1}v = \frac{1}{\Delta t}\int \overline{u_{ey}}^{n}v + \int f_{ey}^{n+1}v \quad \forall v \in G, \\ \displaystyle\frac{1}{\Delta t}\int u_{ez}^{n+1}v - \frac{1}{\epsilon}\int \partial_z\Phi^{n+1}v + \frac{1}{\epsilon}\int u_{ex}^{n+1}v = \frac{1}{\Delta t}\int u_{ez}^{n}v + \int f_{ez}^{n+1}v \quad \forall v \in V_i, \\ \displaystyle\frac{1}{\Delta t}\int u_i^{n+1}\delta + \int \nabla\Phi^{n+1}\delta - \int (u_i^{n+1} \times B)\delta = \frac{1}{\Delta t}\int u_i^{n}\delta + \int f_i^{n+1}\delta \quad \forall \delta \in V. \end{cases}$$

In the numerical implementation, the spaces G and A are not directly available in the MFEM framework, and their construction at the discrete level is not straightforward.

For this reason, we do not explicitly build these spaces. Instead, we impose the restrictions associated with G and A weakly by introducing suitable Lagrange multipliers.

This allows us to enforce the projection constraints defining these spaces directly at the variational level, without requiring an explicit representation of the corresponding subspaces.

The precise construction of these Lagrange multipliers and the resulting discrete formulation will be detailed in the next section.

9.3 Introduction of Lagrange multipliers

The variational formulation obtained above relies on the decomposition

$$V_i = A \oplus G,$$

where A denotes the space of fluctuations and G the space of functions that are constant along the magnetic field lines. Although this decomposition is well suited for the theoretical analysis, its direct implementation requires the construction of finite element spaces adapted to both subspaces A and G.

To simplify the numerical implementation, we enlarge the approximation spaces and allow the unknown u'_{ey} to belong to the full space V_i. In doing so, the condition $u'_{ey} \in A$ is no longer automatically satisfied and must be imposed as a constraint. Since A is characterized as the orthogonal complement of G, this constraint is expressed by

$$\int_\Omega u'_{ey}\, g\, dx = 0, \qquad \forall g \in G.$$

The constraint is enforced weakly by introducing a Lagrange multiplier $l \in G$. The resulting formulation is therefore equivalent to the original one while allowing the use of a single finite element space for the discretization of u'_{ey}.

A second simplification consists in replacing the space G itself by the larger space V_i. In the continuous setting, functions of G satisfy the property

$$\partial_y g = 0.$$

Consequently, the unknowns $\overline{u_{ey}}$ and l, originally belonging to G, are now sought in V_i and their membership in G is recovered through the constraints

$$\partial_y \overline{u_{ey}} = 0, \qquad \partial_y l = 0.$$

These constraints are imposed weakly by introducing two additional Lagrange multipliers λ and μ belonging to

$$L = \left\{ v \in V_i \,;\, \partial_y v \in L^2, \quad v_{|\Gamma_{in}} = 0 \right\}.$$

The final formulation is therefore a mixed saddle-point problem involving the unknowns

$$\left(\Phi, u_{ex}, u'_{ey}, u_{ez}, \overline{u_{ey}}, l, \lambda, \mu, u_i\right).$$

This approach preserves the constraints defining the original spaces while considerably simplifying the finite element implementation. To set every element from A to V we add a condition on the system which is the orthogonality between A and G but for the system to remain correct we need to add a Lagrange multiplier in G. It gives us the following system :

$$
\begin{cases}
\int \partial_x(u_{ix}^{n+1} - u_{ex}^{n+1})v + \int \partial_z(u_{iz}^{n+1} - u_{ez}^{n+1})v + \int \partial_y(u_{iy}^{n+1} - u_{ey}'^{n+1} - \overline{u_{ey}}^{n+1})v = 0 \quad \forall v \in H^1(\Omega), \\
\frac{1}{\Delta t}\int u_{ex}^{n+1}v - \frac{1}{\epsilon}\int \partial_x\Phi^{n+1}v - \frac{1}{\epsilon}\int u_{ez}^{n+1}v = \frac{1}{\Delta t}\int u_{ex}^{n}v + \int f_{ex}^{n+1}v \quad \forall v \in V_i, \\
\frac{1}{\Delta t}\int u_{ey}'^{n+1}v - \frac{1}{\epsilon}\int \partial_y\Phi^{n+1}v + \int l^{n+1}v = \frac{1}{\Delta t}u_{ey}'^{n}v + \int f_{ey}^{n+1}v \quad \forall v \in V_i, \\
\int u_{ey}'^{n+1}v = 0 \quad \forall v \in G, \\
\frac{1}{\Delta t}\int \overline{u_{ey}}^{n+1}v = \frac{1}{\Delta t}\int \overline{u_{ey}}^{n}v + \int f_{ey}^{n+1}v \quad \forall v \in G, \\
\frac{1}{\Delta t}\int u_{ez}^{n+1}v - \frac{1}{\epsilon}\int \partial_z\Phi^{n+1}v + \frac{1}{\epsilon}\int u_{ex}^{n+1}v = \frac{1}{\Delta t}\int u_{ez}^{n}v + \int f_{ez}^{n+1}v \quad \forall v \in V_i, \\
\frac{1}{\Delta t}\int u_{i}^{n+1}\delta + \int \nabla\Phi^{n+1}\delta - \int (u_i^{n+1} \times B)\delta = \frac{1}{\Delta t}\int u_i^n\delta + \int f_i^{n+1}\delta \quad \forall \delta \in V.
\end{cases}
$$

Finally to set the element of G in V_i we introduce two new Lagrange multipliers $\lambda, \mu \in L = \{v \in V_i | \partial_y v \in L^2, v_{|\Gamma_{in}} = 0\}$

$$
\begin{cases}
\int \partial_x(u_{ix}^{n+1} - u_{ex}^{n+1})v + \int \partial_z(u_{iz}^{n+1} - u_{ez}^{n+1})v + \int \partial_y(u_{iy}^{n+1} - u_{ey}'^{n+1} - \overline{u_{ey}}^{n+1})v = 0 \quad \forall v \in H^1(\Omega), \\
\frac{1}{\Delta t}\int u_{ex}^{n+1}v - \frac{1}{\epsilon}\int \partial_x\Phi^{n+1}v - \frac{1}{\epsilon}\int u_{ez}^{n+1}v = \frac{1}{\Delta t}\int u_{ex}^{n}v + \int f_{ex}^{n+1}v \quad \forall v \in V_i, \\
\frac{1}{\Delta t}\int u_{ey}'^{n+1}v - \frac{1}{\epsilon}\int \partial_y\Phi^{n+1}v + \int l^{n+1}v = \frac{1}{\Delta t}u_{ey}'^{n}v + \int f_{ey}^{n+1}v \quad \forall v \in V_i, \\
\int u_{ey}'^{n+1}v + \int \partial_y v\mu^{n+1} = 0 \quad \forall v \in V_i, \\
\frac{1}{\Delta t}\int \overline{u_{ey}}^{n+1}v + \int \partial_y v\lambda^{n+1} = \frac{1}{\Delta t}\int \overline{u_{ey}}^{n}v + \int f_{ey}^{n+1}v \quad \forall v \in V_i, \\
\int \partial_y\overline{u_{ey}}^{n+1}v = 0 \quad \forall v \in L, \\
\int \partial_y l^{n+1}v = 0 \quad \forall v \in L, \\
\frac{1}{\Delta t}\int u_{ez}^{n+1}v - \frac{1}{\epsilon}\int \partial_z\Phi^{n+1}v + \frac{1}{\epsilon}\int u_{ex}^{n+1}v = \frac{1}{\Delta t}\int u_{ez}^{n}v + \int f_{ez}^{n+1}v \quad \forall v \in V_i, \\
\frac{1}{\Delta t}\int u_{i}^{n+1}\delta + \int \nabla\Phi^{n+1}\delta - \int (u_i^{n+1} \times B)\delta = \frac{1}{\Delta t}\int u_i^n\delta + \int f_i^{n+1}\delta \quad \forall \delta \in V.
\end{cases}
$$

We then arrange the system so that the matrix is as symmetric as possible. It gives us the following system of equations :

$$
\begin{cases}
\int \partial_x(u_{ix}^{n+1}-u_{ex}^{n+1})v+\int \partial_z(u_{iz}^{n+1}-u_{ez}^{n+1})v+\int \partial_y(u_{iy}^{n+1}-u_{ey}'^{n+1}-\overline{u_{ey}}^{n+1})v=0 \quad \forall v\in H^1(\Omega),\\
\frac{1}{\Delta t}\int u_{ex}^{n+1}v-\frac{1}{\epsilon}\int \partial_x\Phi^{n+1}v-\frac{1}{\epsilon}\int u_{ez}^{n+1}v=\frac{1}{\Delta t}\int u_{ex}^{n}v+\int f_{ex}^{n+1}v \quad \forall v\in V_i,\\
\frac{1}{\Delta t}\int u_{ey}'^{n+1}v-\frac{1}{\epsilon}\int \partial_y\Phi^{n+1}v+\int l^{n+1}v=\frac{1}{\Delta t}u_{ey}'^{n}v+\int f_{ey}^{n+1}v \quad \forall v\in V_i,\\
\frac{1}{\Delta t}\int u_{ez}^{n+1}v-\frac{1}{\epsilon}\int \partial_z\Phi^{n+1}v+\frac{1}{\epsilon}\int u_{ex}^{n+1}v=\frac{1}{\Delta t}\int u_{ez}^{n}v+\int f_{ez}^{n+1}v \quad \forall v\in V_i,\\
\frac{1}{\Delta t}\int \overline{u_{ey}}^{n+1}v+\int \partial_y v\lambda^{n+1}=\frac{1}{\Delta t}\int \overline{u_{ey}}^{n}v+\int f_{ey}^{n+1}v \quad \forall v\in V_i,\\
\int \partial_y\overline{u_{ey}}^{n+1}v=0 \quad \forall v\in L,\\
\int u_{ey}'^{n+1}v+\int \partial_y v\mu^{n+1}=0 \quad \forall v\in V_i,\\
\int \partial_y l^{n+1}v=0 \quad \forall v\in L,\\
\frac{1}{\Delta t}\int u_i^{n+1}\delta+\int \nabla\Phi^{n+1}\delta-\int (u_i^{n+1}\times B)\delta=\frac{1}{\Delta t}\int u_i^{n}\delta+\int f_i^{n+1}\delta \quad \forall \delta\in V,
\end{cases}
$$

which can then be written as the following linear system:

$$
\begin{pmatrix}
0 & -b_x & -b_y & -b_z & 0 & 0 & 0 & 0 & b\\
-b_x^T & \frac{\epsilon}{\Delta t}M & 0 & -M & 0 & 0 & 0 & 0 & 0\\
-b_y^T & 0 & \frac{\epsilon}{\Delta t}M & 0 & 0 & 0 & \epsilon M & 0 & 0\\
-b_z^T & M & 0 & \frac{\epsilon}{\Delta t}M & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{1}{\Delta t}M & b_y^T & 0 & 0 & 0\\
0 & 0 & 0 & 0 & b_y & 0 & 0 & 0 & 0\\
0 & 0 & M & 0 & 0 & 0 & 0 & b_y^T & 0\\
0 & 0 & 0 & 0 & 0 & 0 & b_y & 0 & 0\\
b^T & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\Delta t}M-C
\end{pmatrix}
\begin{pmatrix}
\Phi^{n+1}\\ u_{ex}^{n+1}\\ u_{ey}'^{n+1}\\ u_{ez}^{n+1}\\ \overline{u_{ey}}^{n+1}\\ \lambda^{n+1}\\ l^{n+1}\\ \mu^{n+1}\\ u_i^{n+1}
\end{pmatrix}
$$

$$
=\begin{pmatrix}
0\\
\frac{\epsilon}{\Delta t}Mu_{ex}^{n}+\epsilon Mf_{ex}^{n+1}\\
\frac{\epsilon}{\Delta t}Mu_{ey}'^{n}+\epsilon Mf_{ey}^{n+1}\\
\frac{\epsilon}{\Delta t}Mu_{ez}^{n}+\epsilon Mf_{ez}^{n+1}\\
\frac{1}{\Delta t}M\overline{u_{ey}}^{n}+Mf_{ey}^{n+1}\\
0\\0\\0\\
\frac{1}{\Delta t}Mu_i^{n}+Mf_i^{n+1}
\end{pmatrix}.
$$

9.4 Numerical results

We consider the following source terms:

$$\mathbf{f}_e(x,y,z,t) = \begin{pmatrix} e^t\cos(\pi x) - \frac{1}{\epsilon}\pi\cos(t)\cos(\pi x)\cos(2\pi y)\sin(\pi z) - \frac{1}{\epsilon}e^t\sin(\pi z) \\ e^t\cos(2\pi y) + \frac{2}{\epsilon}\pi\cos(t)\sin(\pi x)\sin(2\pi y)\sin(\pi z) \\ e^t\sin(\pi z) - \frac{1}{\epsilon}\pi\cos(t)\sin(\pi x)\cos(2\pi y)\cos(\pi z) + \frac{1}{\epsilon}e^t\cos(\pi x) - \frac{1}{\epsilon}\sin(2\pi z) \end{pmatrix},$$

$$\mathbf{f}_i(x,y,z,t) = \begin{pmatrix} e^t\cos(\pi x) + \pi\cos(t)\cos(\pi x)\cos(2\pi y)\sin(\pi z) + e^t\sin(\pi z) \\ e^t\cos(2\pi y) - 2\pi\cos(t)\sin(\pi x)\sin(2\pi y)\sin(\pi z) \\ e^t\sin(\pi z) + \pi\cos(t)\sin(\pi x)\cos(2\pi y)\cos(\pi z) - e^t\cos(\pi x) \end{pmatrix}.$$

And the associated exact solutions :

$$\phi(x,y,z,t) = \cos(t)\ \cos(2\pi y)\ \sin(\pi x)\ \sin(\pi z),$$

$$\mathbf{u}_i(x,y,z,t) = \begin{pmatrix} e^t\cos(\pi x) \\ e^t\cos(2\pi y) \\ e^t\sin(\pi z) \end{pmatrix}, \quad \mathbf{u}_e(x,y,z,t) = \begin{pmatrix} e^t\cos(\pi x) - \sin(2\pi z) \\ e^t\cos(2\pi y) \\ e^t\sin(\pi z) \end{pmatrix}.$$

We computed the L^2 error convergence for different values of μ and Δx, namely $\mu = 1$, $\mu = 10^{-3}$ and $\mu = 0$, using mesh sizes $\Delta x = 0.25$, 0.1667, 0.1 and 0.0625. Since the computational cost becomes significant for small values of μ, a solver tolerance of 4.5×10^{-6} was adopted. Additional tests showed that reducing the tolerance further led to a slight improvement in the computed L^2 errors, but at the expense of a substantial increase in computational time. The chosen tolerance therefore represents a reasonable compromise between accuracy and efficiency.

The objective of these experiments was primarily to verify the correctness and robustness of the proposed formulation. In particular, we aimed to assess whether the method exhibits the expected convergence behavior and to investigate the influence of μ on the number of iterations required by the linear solver. For these tests, the velocity variables were approximated using piecewise constant finite elements (order 0) in the L^2 space, while the potential was approximated using piecewise linear finite elements (order 1) in the H^1 space. In all simulations, the time step was chosen as $\Delta t = \Delta x$ in order to keep the temporal and spatial discretization errors at comparable levels.

The obtained results are presented below.

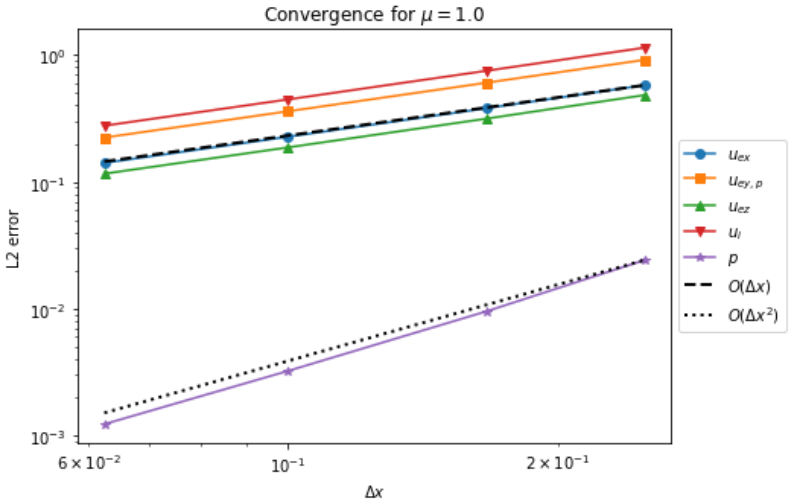


Figure 9: L^2 error convergence for $\mu = 1$.

Figure 9 shows that for $\mu = 1$, the numerical results exhibit a clear convergence trend as the mesh is refined. The errors associated with the velocity components remain close to the reference first-order slope $O(\Delta x)$, while the error corresponding to p decreases significantly faster, with a convergence rate close to $O(\Delta x^2)$. This behavior is consistent with the finite element spaces used in the discretization: piecewise constant approximations for the velocity variables and a piecewise linear approximation for the potential. Interestingly, the convergence rate observed for the potential is slightly higher than the theoretical rate expected from the approximation order of the finite element space. Such behavior is commonly observed for sufficiently smooth manufactured solutions and moderate mesh refinements. Overall, these results confirm that the mixed formulation is correctly implemented and that the expected convergence properties are recovered in the non-stiff regime.

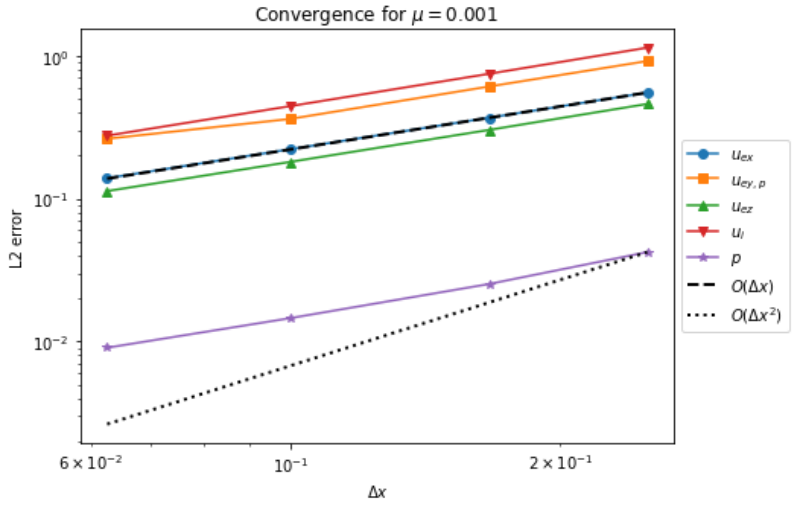


Figure 10: L^2 error convergence for $\mu = 10^{-3}$.

Figure 10 shows that for $\mu = 10^{-3}$, the convergence behavior remains very similar to that observed for $\mu = 1$. The velocity errors decrease approximately with first-order accuracy, while the potential exhibits a slower convergence rate than in the previous case, although the observed convergence behavior remains satisfactory. The fact that reducing μ by three orders of magnitude does not significantly deteriorate the convergence properties indicates that the proposed formulation remains stable in a moderately stiff regime. This suggests that the introduction of the Lagrange multipliers successfully preserves the consistency of the discretization while handling the additional constraints.

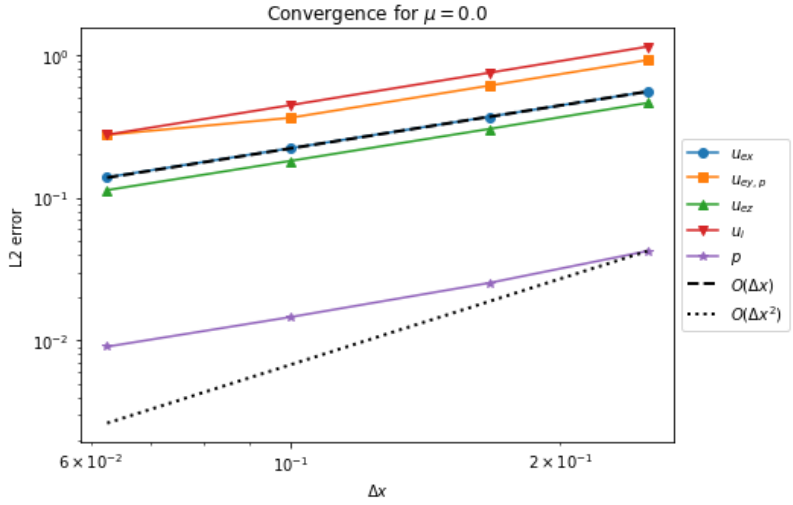


Figure 11: L^2 error convergence for $\mu = 0$.

Figure 11 shows that for the limit case $\mu = 0$ corresponds to the asymptotic regime targeted by the proposed method. The numerical results show convergence rates comparable to those obtained for $\mu = 10^{-3}$. In particular, both the velocity variables and the potential variable continue to exhibit approximately first-order convergence. The absence of any significant loss of accuracy in the singular limit demonstrates that the discretization remains robust when μ tends to zero. This behavior provides encouraging evidence that the proposed formulation captures the correct asymptotic limit and is consistent with the objectives of an asymptotic-preserving approach.

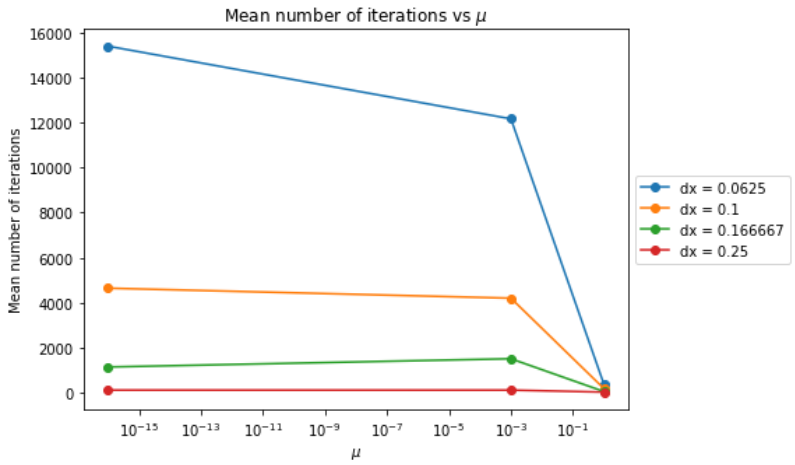


Figure 12: Average number of solver iterations as a function of μ for different mesh sizes.

Figure 12 presents the average number of iterations required by the linear solver as a function of μ for different mesh sizes. For coarse meshes, the iteration count remains relatively moderate and only weakly dependent on μ. However, for the finest mesh, a significant increase in the number of iterations is observed as μ decreases. This behavior reflects the increasing stiffness of the system in the asymptotic regime and highlights the growing difficulty of solving the associated linear systems. Nevertheless, convergence is still achieved for all tested values of μ, demonstrating the robustness of the proposed formulation. These results also indicate that the development of dedicated preconditioning strategies could substantially improve the computational efficiency of the method for very small values of μ.

10 Conclusion

The objective of this work was to investigate the numerical approximation of a reduced electromagnetic subsystem arising from the two-fluid Braginskii equations and to prepare the implementation of an asymptotic-preserving strategy within the finite element library MFEM.

A significant part of the project was devoted to understanding the mathematical and numerical tools required for this objective. To this end, several model problems of increasing complexity were implemented and studied. These test cases made it possible to validate the finite element discretization, the treatment of mixed formulations, and the implementation of implicit time integration schemes. In each case, the numerical results reproduced the expected convergence rates, providing confidence in the correctness of the implementation.

Based on the work of Pierre Degond [6], a reformulation of the reduced electromagnetic model was introduced. The main difficulty arises from the presence of constraints associated with the decomposition of the solution into components parallel and orthogonal to the magnetic field. To simplify the finite element implementation, these constraints were enforced using Lagrange multipliers. This approach preserves the mathematical structure of the original problem while allowing the use of standard finite element spaces.

The numerical experiments performed on the reformulated system demonstrated stable and consistent convergence properties for several values of the parameter μ, including the limiting case $\mu = 0$. The results indicate that the method remains accurate as the stiffness of the problem increases, which is an encouraging first step toward the construction of an asymptotic-preserving scheme. In addition, the study of the linear solver iterations showed that the formulation remains solvable in the asymptotic regime, although the computational cost increases for small values of μ and fine meshes.

Several directions remain open for future work. First, a more complete theoretical analysis of the mixed formulation should be carried out. Second, the development of efficient preconditioners would be essential to improve the performance of the linear solver in highly stiff regimes. Finally, the methodology developed here should be extended to the full reduced electromagnetic model and ultimately to the complete two-fluid Braginskii system.

Overall, this work provides a solid numerical and implementation framework for future developments toward asymptotic-preserving finite element methods for plasma simulations.

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